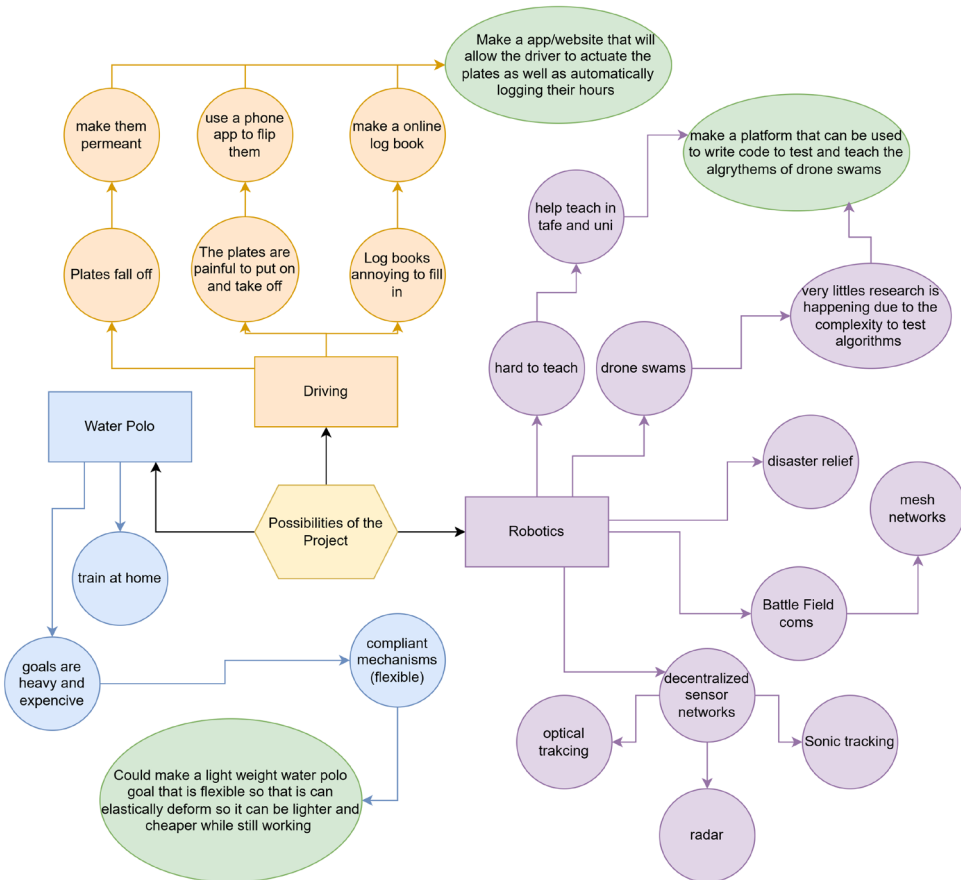
Project Proposal

Identification and Exploration of the Need: Exploration of Needs

Initial Mind Map: Several ideas were considered when selecting an option to pursue for my major design project.



Initial Ideas

Idea 1: The first idea was a drone swarm that can be used for teaching swarming technologies to students. The goal is to provide scientists and students with the hardware and software platform to develop new algorithms, as there is a lack of development in the field of swarms.



Ongoing Evaluation: The reason I did not choose this project was because it is incredibly complex and hence I would not be able to make a polished final project.

Idea 2: The next idea was to create a water polo goal, as currently water polo goals are very heavy and expensive. They are designed this way so when people shoot at the goal with great force it can withstand it. To solve this problem, I was going to make a goal that is flexible whereby the soft material could move on impact and then recover to its original shape.



The reason I did not choose this project was because it would be difficult to iterate and test new designs as each version would require a new goal, which is a significant time investment.

Idea 3: The final option and the project I decided to go with was an automatic L-plate. This L-plate solves the need that L-plates and P plates regularly fall off, leading to environmental pollution through the creation of litter and also great annoyance for the public. In addition, the ongoing cost of replacement. My solution would also solve the tedious process of putting on L-plates and taking them off as well as filling in the log book through an app I would build with it. People often drive with the wrong plates due to laziness, which is both illegal and dangerous. I have lots of experience making electronics projects and designing 3D printed parts but I want to learn how to optimise mass production through 3D printing as this is a useful skill for the future. This project will also help me learn about software design, app development, microcontrollers, and GPS.

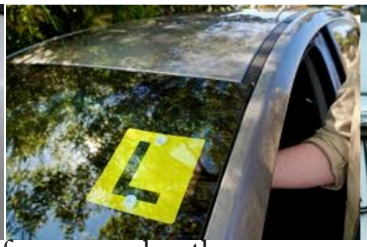
The reason I did not choose this project was because it is incredibly complex and hence I would not be able to make a polished final project.

Project Proposal

Existing Products









It is difficult to determine existing products for this type of a proposal as there are none. The only functioning existing product that exists at the moment are physical L plates that need to be stuck on manually.

Why is it a need and what problem does it solve?

The project is needed because L-plates are annoying to take on and off and they often fall off, causing litter. The increased convenience and better mounting design will result in fewer people breaking the law by driving without the proper plates, and improve the accurate filling in of the logbook.

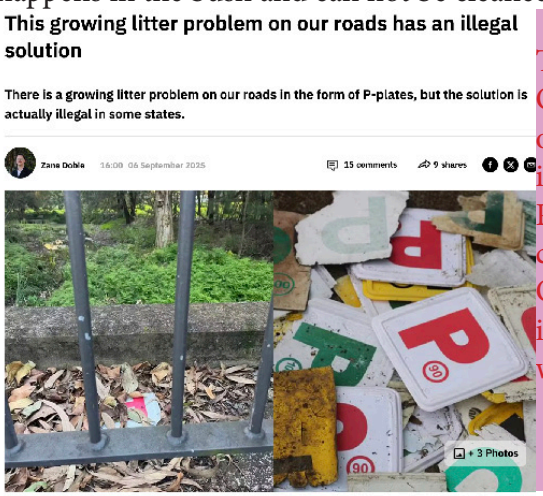
- Not filled out log books
- Sporadic driving practise → worse drivers
- Driving with incorrect plates
- Plates falling off → litter and annoyance
- Plastic Waste
- Inconvenience of removing and adding the plate to fit multiple drivers

Secondary Research

Plastic Waste

• Poor fixtures: L and P plates are required to be displayed outside of the car which can result in them getting damaged quicker. The clip/holder that are typically provided with the L and P plates are also generally made of poor quality plastic, adding to the waste.

• Non-Recycled Material: L plates create plastic waste due to the non-recyclable materials they use. So when they fall off on the street, they cause litter and pollution resulting in added issues to the environment especially if this happens in the bush and can not be cleaned up


This growing litter problem on our roads has an illegal solution

There is a growing litter problem on our roads in the form of P-plates, but the solution is actually illegal in some states.

Zane Doble 16:00 06 September 2025 15 comments 9 shares

Photos: Facebook

This article unpacked how widespread the issue of discarded L plates is. One of the key findings is that there were 101 L plates found on the side of the road along a 1 km stretch. This demonstrates that this is a systemic issue and license plates are a large source of plastic pollution and litter. Plastic pollution and is harmful to the environment as it is a primary contributer to global warm, a prevalent issue in current society.

One thing that the article recommends to do is to mount the L plates internally this supports my idea about mounting them internally as that is what I was intending to do.

Website: https://www.drive.com.au/caradvice/this-growing-litter-problem-on-our-roads-has-an-illegal-solution/

This article unpacks how Australia has a growing pollution problem and how our recycling infrastructure cannot keep up with this. In 2005 we had a 14 .7% recycling success rate whereas now it has dropped to 14% in 2024. This is due to a lack of sufficient infrastructure and the fact that we are exceedingly reliant on plastic in Australia. The government intends to enact stricter laws regulating how packaging can be sold in products, for example banning use of soft plastic bags.

They are also investing more in the recycling sector and information campaigns to inform the public about recycling practices.

My protect will reduce the amount of plastic waste therefore reducing the load on the recycling infrastructure of Australia


Australia's amount of plastic waste surges as recycling rates fail to improve

Australians produced 3.2m tonnes of plastic waste in 2023-24, up from 3m tonnes the previous year

Get our breaking news email, free app or daily news podcast

The total amount of plastic waste has doubled since 2005 (from 1.57m tonnes) but recycling rates have largely stalled despite investments in increased capacity. Photograph: Franc

Website: https://www.theguardian.com/environment/2025/aug/24/australias-amount-of-plastic-waste-surges-as-recycling-rates-fail-to-improve

Project Proposal

Secondary Research

Plastic Waste



This article finds that there are large quantities of micro plastics in Australia Australian waterways. It conducted a survey in the Illawarra river and it found that the levels of micro plastics they were comparable to China. This means Australia is suffering from a severe environmental problem. You can see from my images below in my primary research that a lot of these L plates have started to breakdown into micro plastics so my solution will reduce the prevalence of these micro plastics. Also micro plastics while also being an environmental issue is a issue to society as they are very unhealthy.

https://www.abc.net.au/news/2026-03-16/nsw-estuaries-microplastics-report/106458862

The New South Wales government is investing $1.25 million into finding solutions for plastic waste. This proves that there are a legislative desire for my product as my product will also reduce the amount of plastic waste not solve it as is done in these cases but it will reduce it.



https://www.nsw.gov.au/ministerial-releases/new-research-funded-to-find-plastic-waste-solutions

Boomerang Alliance



State/Territory	External only	Internal assumed	Not specified	Materials
Western Australia			x	Printable
South Australia		x	x	See-through plates for sale
Northern Territory			x	
Tasmania			x	
Victoria		x		
New South Wales	x			Plastic freebies
Australian Capital Territory			x	
Queensland			x	Downloadable template

This proves that P plate pollution is a large problem and that putting them inside is legal in some states. But this organisation is trying to lobby the government to allow it to be more widespread.

https://boomerangalliance.org.au/pplate

Identification and Exploration of the Need: Exploration of Needs





Although this is a good proposed solution, it still would not acknowledge the following issues

- Visibility: Many of the cars are tinted and it may also not be visible during the day due to the sunlight reflection
- Thermal: Cars can get very hot, up to 40-50 degrees celsius in the summer, which could result in the plastic melting
- Blocking vision for the drivers through rear window

How would my product improve user experience over existing or similar designs?

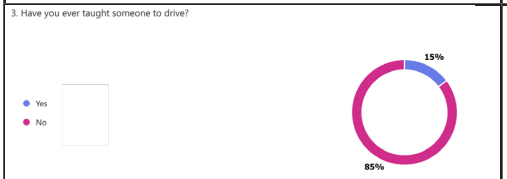
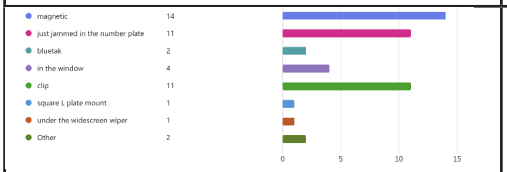
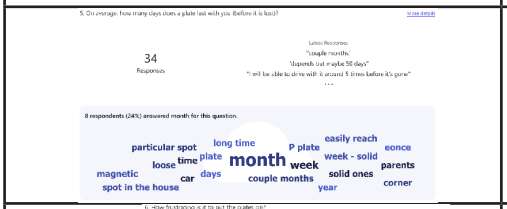
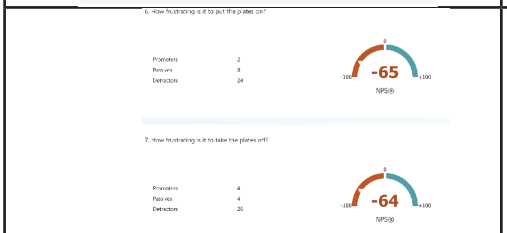
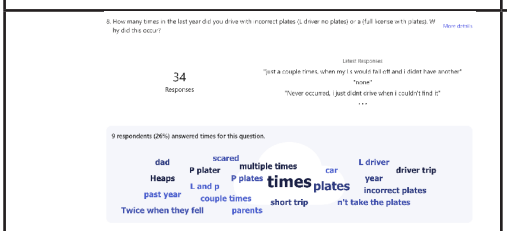
With fixed plates that can be electronically displayed and hidden my design removes the arduous and time consuming process of walking around to the front and back of the car to put L-plates on. Often learner drivers miss out on driving trips when passengers are rushed which reduces opportunities to develop their driving skills. The device will securely hold the plates to reduce likelihood of them falling off and littering the road.

Project Proposal

Point of Difference

Existing products are all a plate that is made of plastic or the black magnetic material that are attached to the vehicle by magnetism, a clip, suction, or adhesion (bluetick or tape). These mounting options all have their strengths and weaknesses but the best option is the clip because it is easier to take on and off and it is quite reliable. My design solution will stay permanently attached to the car and will use an electronic system to toggle the plate on and off. This electronic system will move a flap or similar mechanical system to controllably reveal and hide the L-plate, thus resulting in the effect of the plate becoming visible and not. This is quicker and easier than the manual systems. The app/website that I will build will allow the user to log hours easily and quickly.

Primary Research

	This defines the scope of the problem by illustrating the reality that these plates are lost fairly regularly hence the current system need, improvement.
	Demonstrates the log books are not filled in correctly. This shows that a solution is necessary to encourage people to fill in their logbook correctly complying with the legal requirements.
	shows that people falsify records out of laziness and other factors such as time constraints. This data provides evidence that people are not acting within the regulations of the laws, and then, consolidating the concept that users inherit laziness leading to danger.
	This data demonstrates that uses typically lose their plates every month or more. This is very frequently! This proves that L plates falling off is a key issue that my target market faces.
	The response to this question indicate that it is very frustrating putting our plates on and off the car. This difficulty primarily leads people to not doing it or forgetting and being lazy about how they mount it as they want to just drive instead of dealing with their plates so they will use poor mounting methods this results in them falling off and causing litter and more annoyance.
	This Question shows that learner drivers will regularly drive without our plates or their parents will forget to take them off this leads to the law getting broken regularly and unintentionally. This is bad because learner drivers need to have L plates for a reason so I can warn other drivers to be careful and if they drive without this it is dangerous to them.

This department research I've conducted Strongly supports my MDP as it proves that there is a need and it doesn't need that. I believe I can solve so this survey I did establishes that I can continue moving forward because I am actually addressing in need.

Project Proposal
Primary Research










I went for a bike ride with my mum and on the way home we were talking about my major work and I started noticing P plates. Once I pointed it out we couldn't stop spotting them every 50 to 100 m. We would stop and see one and take a photo of it. Here are some of the photos that we found.
OVERALL WE FOUND 50 P PLATES IN A STRETCH OF 7 KM
That is a lot of paper plates and a lot of plastic waste on the side of the road. A Qualitative observation I made was that the plastic waste was primarily License plates.
This is likely due to P platers laziness at not mounting them properly.

This primary research makes me feel very confident that my project is solving a real need. Not only that, but I also saw a car that had crashed and I thought to myself what if that was a P plater that hadn't done all their hours. Both of these together help me reaffirm my assurance that this project could benefit society the environment and the individual drivers.

Project Proposal

Design Brief

For my major design project, I plan to design an automatic L-plate. From my research and brainstorming, I have discovered a problem that I can solve which is the bad practise around displaying L-plates. Using my experience of CAD modelling and designing for 3D printing as well as my experience in electronics design and microcontrollers gained from personal projects and design technology, this give me an advantage when making this product. Using my skills, I intend to make an automatic L-plate this L-plate should allow the user to go from displaying them to not. This will be controlled form an app or a website on the user's phone a integrated log book that will be filled out automatically will be integrated into it.

Design Situation

The project is needed because L-plates are annoying to take on and off and they often fall off, causing litter. The increased convenience and better mounting design will result in fewer people breaking the law by driving without the proper plates, and improve the accurate filling in of the logbook.

Not filled out log books

Sporadic driving practise => worse drivers

Driving with incorrect plates

Plates falling off => litter and annoyance

How does it improve the user experience over existing or similar design solutions?

With fixed plates that can be electronically displayed and hidden my design removes the arduous and time consuming process of walking around to the front and back of the car to put L-plates on. Often learner drivers miss out on driving trips when passengers are rushed which reduces opportunities to develop their driving skills. The device will securely hold the plates to reduce likelihood of them falling off and littering the road.

Target Market

Non-fully licensed drivers and anyone who shares a car with them. This includes the L-plate divers leaning in parents cars and P-plate drivers who share with their parents.

Where do they live?

In every state of Australia.

Why will it appeal to the user?

This will appeal to the user because it is better than the other options in the following ways

Less litter from dropped plates and don't have to keep buying them

Integrated logbook filling out

Easier to switch from displaying plate to not (don't have to get out of car)

Helps you become a better driver because of the ai teacher built into the app

I'm very well suited to serving this target market because I am a member of the target market which means that I can ask myself what I would like in a product which means that I can more effectively design this product to suit my target market because I know exactly what they want.
In addition to this I have easy access to my target market as they are my peers and classmates. This lets me in quickly taking advantage of rapid that the FDM process offers.

Project Proposal

Areas of Investigation



This chart allowed me to quite clearly unpacked the things that I Should investigate if I am to build a good product. As there is quite a large amount of scope, this will take quite a significant amount of time to learn about. But once I have good general knowledge in these spaces I can then start doing specific research that is relevant to the things that I want to do. I should first learn about general 3-D printing good aesthetic design and then learn specifically about things needed for my project such as printing in place mechanisms.

Project Proposal
Area of Investigation

Area of Investigation	Justification	What will be investigated	How it will be investigated
Where can I legally put a L plate?	Because it determines my mounting options	I will ask a llm to find me the relevant legislation, then I will read that	Very as legality is a highly important design parameter.
This entire project hinges on whether or not I am legally allowed to do it and make it. If this product is illegal my target market and my entire solution would become irrelevant as you would be breaking the laws I was trying to stop you breaking by using my product.			
How to design print in place?	It will enable me to make a more reliable, stronger and easier to print	I will watch YouTube videos to understand little tips ect. look on places like thangs to gain inspiration from other designs.	I will ask both my mentor and also need to do lots of prototyping as this it is art not a science.
Designing print in place is incredibly useful as it means that I can print mechanisms that I couldn't normally do as they're impossible to manufacture traditionally. So learning this skill would allow me to have a superior product then could be manufactured with something like injection moulding for example.			
The strengths and weaknesses of different 3D printed filaments	As different filaments are better and worse at different things for example, PLA is easy to print deforms in a hot car and also suffers from UV degradation.	I will do research on the YouTube channels “slant 3D”, “CNC kitchen”, “zach freedmen’s” In addition, I will conduct primary research by printing the my prototypes and laving in oven at 70 C as this is 5 C hotter then a hot car in summer. I will conduct a strength test by mounting it under my car and driving on a gravel road. I will also leave the a prototype in the sun for a large period of time then do a visual assessment.	I will look online watch videos comparing them I may ask on a forum I will cross check with an AI model. My mentor also has lots of experience in this area so I may want to ask him.
Every single filament is made up of different plastic polymers. They all have their own strengths and weaknesses for example PLA degrades under UV light whereas petg does not but petg does not print overhangs as well. Once I determined what filament I'm going to use it allows me to start dialling in the exact design but until then I can't optimise for 3D printing fully.			

Project Proposal
Area of Investigation

What actuator is best suited for my design	There are many actuators on the market, from very large ones to really small ones. I need one with enough torque, waterproof, closed loop (this means that it knows where it is) I also want to be able to read the position of the actuator to verify that the plate actually moved.	I will describe to a llm what the task I need to achieve is with a prompt something like "I am make a L plate that actuates I want something the can produce a relatively small amount to torque I also need to read the position f the actuator with a micro controller whether be a second sensor or be inbuilt can you suggest various low cost weather proof actuators that suite" Then I will search up the suggested actuator to look for images and cost I will make a table ranking then each out of 10 for cost, strength, ease of integration, and durability. Finaly I will test the top 3	
There are different sizes and strength of actuators some of them are louder some of them are stronger some of them are heavier some of them are bigger. Once I've established what actuator I need to get and use I can then start designing the case for it but only once I've determined and decided on my actuator until then I can't proceed.			

Project Proposal
Area of Investigation

Learn to make a app how to program	Because I want to have a app to control the L plate	I am doing Software design development as one of my HSC subjects and I was taught the relevant information in the curriculum that build me to make this app. In summary I frequently looked on websites like stack overflow and geeks for geeks as well as asking a llm to help solve issue. Most projects have documentation that explain how to use and write in that programming language I read that that was my primary source of information.	High as I need to be able to program to make my app so then I can make a complete working product
Given that I want to build an app to work with the hardware I need to know how to build an app there are different types of apps and some of them have different levels of complexity and given that I'm a year 12 student some of them will not be in the scope of what I can attain so this means I have to have realistic expectations			
How to design for 3D Printing	I intended to 3D to print my product as 3D printing can be mass produced as well as being accessible to a DT student. When designing for 3D printing you can make 'impossible' geometry this gives you advantages I you know how to do it.	There's lots of good information available on the Internet. And the 3D printing community is quite friendly and willing to answer questions.	I will have discussions with my mental and I will also look at the plentiful knowledge online by watching videos and reading blogs
Given that it is an absolute bottomless pit optimizing for printing I expect I will be doing a large amount of prototyping and iterations as that is one of the only ways to get good at this skill there's both by practicing it and by every single time you print something you learn what you need to do to improve it for the next one			

Project Proposal

Criteria to Evaluate Success

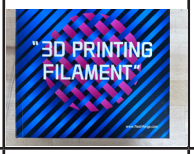
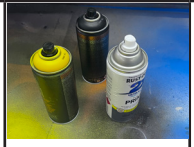
Criteria	Relevance	Implantation
Function - Ergonomics: Easy to take on and off - Quick to put on and off	The L Plate must flip open, shut and log hours fastrer than it is to do it manually. The time it takes a typical leaner driver to put my plates on is 46 seconds Displaying L-plates (and taking them down) must be significantly more convenient than the old way is because the user won't use it. Log hours must not take longer than manual and deliver additional benefits such as permanent record keeping (indelibility) The product will not be used, and therefore the need won't be solved	The L-plate should take 50 seconds to toggle and the log book component should take 48 seconds to complete I will be testing each prototype and the rate of which it takes to put on and off and also the ease of it. If there are any obstructions I will be improving this from prototype to prototype This also needs to be easy to use. So I will be evaluating the ease of the use of this product. I will also be surveying the demographic on
Aesthetics - Colour that does not stand out - Design should be sleek as it should go onto any car	The plate must blend into the cars aesthetic so that it is not an eyesore. f the plate looks ugly and does not blend in to the cars aesthetics, some users won't want to use it. It should be a pleasing design to look at and should not stand out too much on the vehicle. The impact if not achieved it the plate will have less adoption as well as not solve the need as well.This will result in people reverting back to the previous solution of the L and P plastic. The L-plate should not be visible foreign 10 meters away, and it should not have exposed wires. The colour should be neutral.	The L-plate should not be visible foreign 10 meters away, and it should not have exposed wires. The colour should be neutral.
Legality - Follows the NSW guidelines	Because this is a legally mandated signage, it must display the L-Plate in a way that complies with the regulations The entire MDP would fail because the plate would not comply with the legal requirements expected from the learner driver and their supervisor.	There is a minimum that is complying with the law or not it is black and white this is the linchpin of the whole project if it is not legal, the entire project is a failure I will follow the NSW and ACT guidelines and plate usage on vehicles closely referring to this website: https://www.service.nsw.gov.au/services/number-plates

Project Proposal

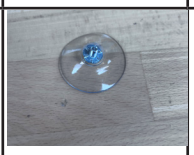
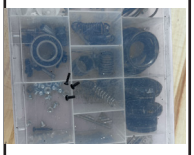
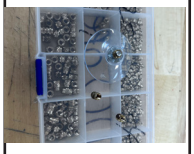
Criteria to Evaluate Success continued...

Criteria	Relevance	Implantation
Quality - Life Cycle - Can sustain wind and other road conditions	The plate should feel high-quality and be high quality. Because this is a 3D printed there is a mindset that all 3D printed products possess inferior quality. However, this is not always true. It is easier to make premium products if by taking full advantage of the 3d printing quality. If it does not feel high quality, people who care about their cars will not put it on the it. If the product is not high quality, it will not work well and break easily thus not solving the need of the plates falling off The impact of a inferior product will result in a smaller portion of the target market being addressed	There must be proper-looking packaging and no exposed wires The product must work reliably and have longevity. Testing will occur with the final prototypes of actually driving with the device attached to the car to ensure that it lasts. The material that the product is made of would also reflect a stronger material than the current solution.
Safety - Safe use for consumers - Should not damage the car	The product needs to be safe to both the consumer and the car itself. The consumers should not experience and cuts while attaching the product on the car The car itself should not get damaged when using the product. This includes paint scratches or dents onto the actual vehicle due to the product	I will test each prototype to ensure that it does not damage the car. A stress test will also take place to determine if there are any impact on both the consumer or the vehicle.
Cost	The L plate must be a reasonable cost. It should not cost than more units that one would spend at the moment. A physical pack of L plates typically costs between $3 and $15 AUD, depending on the attachment style. Basic Plastic / Tie-on Plates: Around $3 – $5 per pack.Magnetic Plates: Around $7 – $10 per pack. Clip-on / Premium Holders: Around $12 – $15 per pack	I will closely evaluate the cost of my experimentation and resources used throughout the process to ensure the product uses materials that do not exceed the concurrent cost.

Project Proposal
Finance Plan

Item	Image	Proposed Cost	Actual Cost	Evaluation
Soldering wire with flux		$30	$0	This particular soldering flux is lead free. This is important as lead free solder is much safer to work with as lead is toxic. It is also of a small diameter which makes it ideal for microelectronics
Printing filament PETG PLA		$25x3	$0	This printing filament was provided by the School DT department they had PLA and PETG in different colours
CA glue (Superglue)		$2	$0	This is superglue. I used it to adhere the acrylic to the 3D printed plastic although in the final product I did not need to use this as I had Designed my solution to be reusable and repairable
Small cable ties		$5	$0	These are small cable ties. I used them to fix the servo horn to the 3D printed part. Initially, I used superglue then I used larger cable ties finally I used small cable ties. I settled on this as it would suit my design better
Box cutter		$17	$0	This Knife is really useful as it allows me to cut the support material away from my printed parts and also to cut wires as the School does not have wire cutters.
9g Servo		$6x10	$5.70x10 $57	These are very common in small electronics projects. They are a small server motor. This is the actuator that I can control via a PWM signal (a way of commanding a position of the actuator) This then allows me to control the position of my louvre. The reason I had to buy so many of them is I broke a couple accidentally
Bambu Lab P1S		$1 200	$0	My Mentor had this printer and it is a really good printer. It prints it very high speed with very hyper precisions and excellent service finish. This allowed me to rapidly iterate prototypes and I also got to learn how to use it
Spray booth		$12 00	$0	This is a spray booth which lets me safely spray paint. I needed it so that I could test whether spray painting was a valid option for applying the L to the plate
Spray paint		$20x3	$0	These are spray paint and primer. I needed them to spray paint. They also need to be the correct colours. The School didn't have all the colour so I tested without the colours and I established that it was not a viable approach.
Contact		$8x3	$8x3 $24	This is contact (the same thing you used to cover books with just different colour). I needed them in the colours of white red and black and it is surprisingly difficult to find. I had to go to 5 different stores to try and get it.

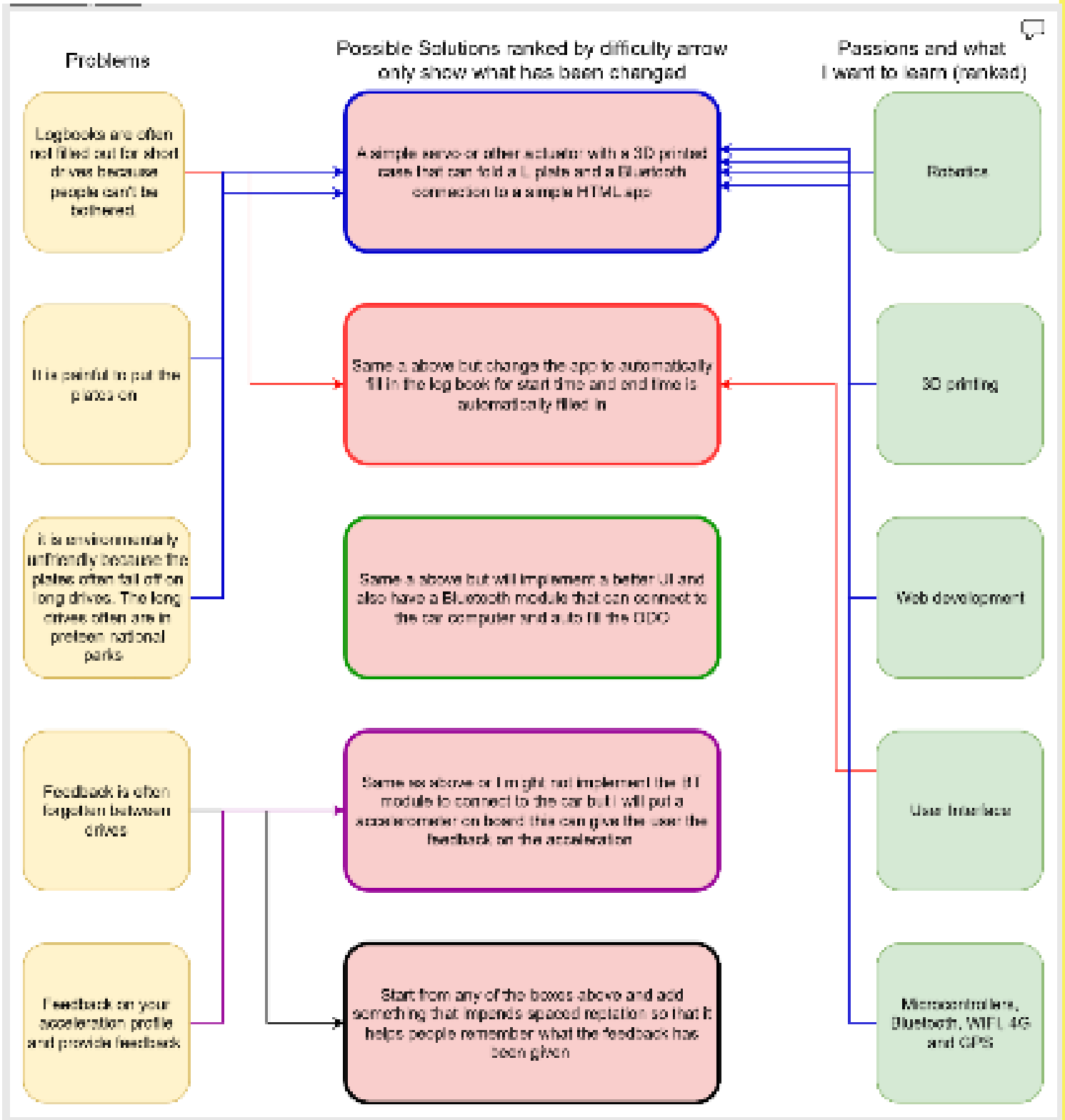
Project Proposal
Finance Plan

Item	Image	Proposed Cost (AUD)	Actual Cost (AUD)	Evaluation
Acrylic		$20	$0	This is acrylic. I initially used it for the base plate but my final design did not include it as it would make it more repairable and remove the need of planned obsolescence.
Band saw		$3840	$0	This is the band saw that I used to cut the acrylic. I did not end up meaning to use it but it was used in prototyping.
Heat Gun		$600	$0	Due to inconsistencies in the manufacturing process of server horns, they were of inconsistent size so I chose to use the heat gun to soften the 3D printed plastic so that they could fit
Wire		$15	$0	This is silicon shielded wire 30AWG wire this is good as it is thin and easy to wrap and the silicon coating means it's flexible
Suction cup		$20	$10.99	These suction cups were a very good fined as they let me successfully mount my L plate to the car. Also they're cheaper than I predicted which was nice.
18650 Protected cell		$15x6	$15x6 90	This is a 18650 protective cell. If it was not a protected cell, my micro controller could drain the battery past where it is safe to potentially causing a fire.
ESP 32 C3		$15x6	$10.65x6 $63.90	This ESP C3 is a common low power low cost micro controller. It has good SDK meaning that it is easy to program on. It is very popular and common so that has lots of example code and forms where you can get help. It also has an integrated battery charger this means I can reduce my count thus reducing cost.
TS101 Soldering iron		$100-150	$0	This is a very good soldering iron. It is powered off a USB C port so that you do not need to buy second power supply meaning that has high power capabilities while also not being bulky and heavy and low cost.
Self tapping screws		$5	$0	These allow me to fix the modules case in position. I chose self tapping screws as they were cheap easy to use and as it was not a load-bearing part it was sufficiently strong.
Threaded inserts		$10	$0	These are threaded inserts. They are cheap while also being very easy to use. You need to heat up a soldering iron and then press them into a hole. This allows you to have a metal thread in plastic part. I have these at home as I have used them in personal projects before.

Hand draw your finance plan or do something to make it look like you actually did it

Project Proposal

Action Plan



Because this major work is quite ambitious I have structured the project in such a way that depending on how my time management goes I can stop at various checkpoints along the development of the project and still have a finished prototype.

This plan also quite effectively outlines the different areas that I can pursue and each of them are mostly decoded from each other meaning that I can work on one without negatively affecting others so it lets me modularise the time I spend so that I can become better at each of these areas separately while working on them and it means that my overall skillet can be lower

An additional benefit is that I can find different mentors for each one of these areas and this means that I can get lots of help from experts in their areas.

Project Development & Realisation

Existing Designs & Solutions

Image:	Plus:	Minus:	Interesting:

Image:	Plus:	Minus:	Interesting:

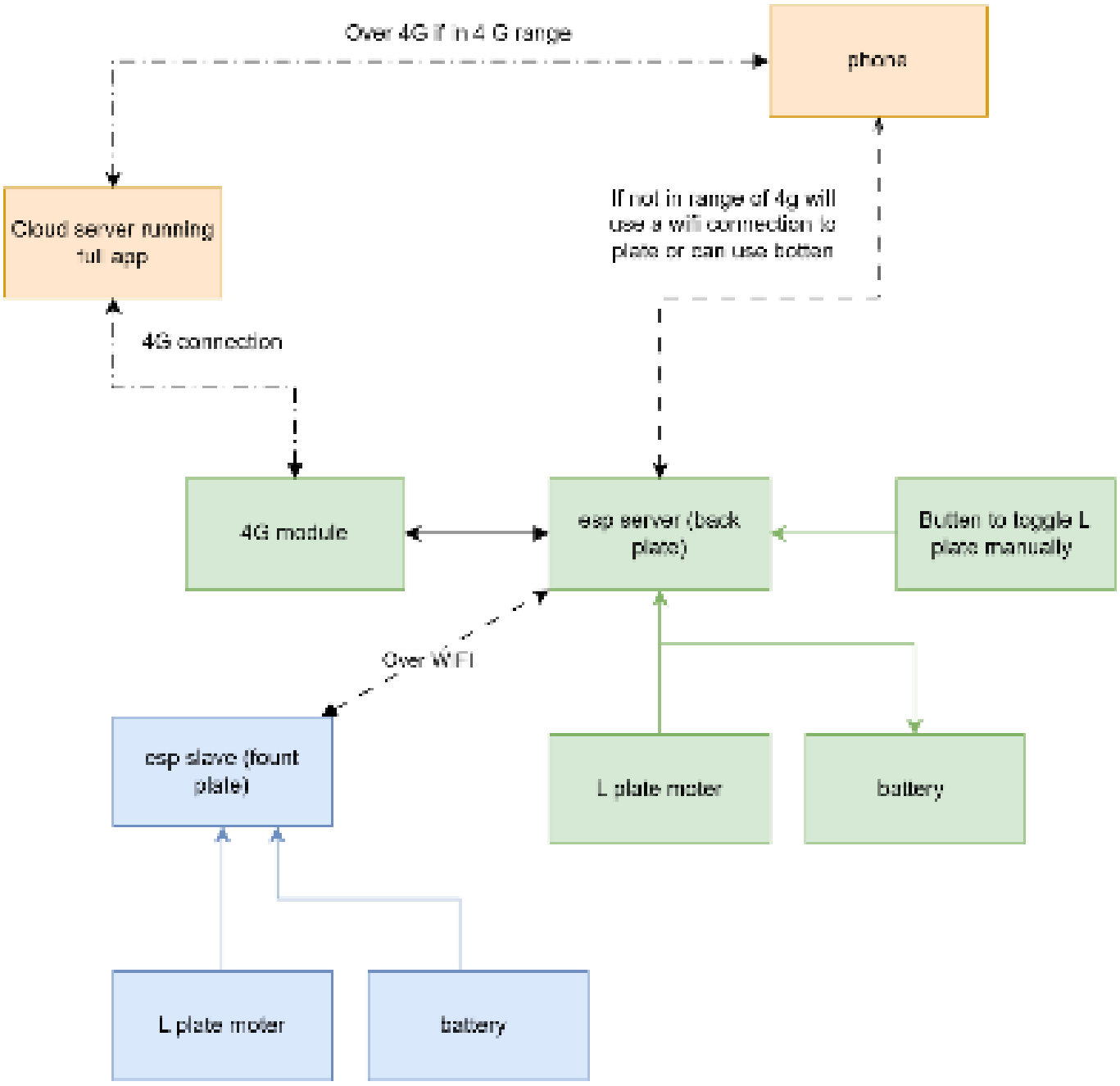
Image:	Plus:	Minus:	Interesting:

Project Development & Realisation

Sketches

Project Development & Realisation

Idea Generation



This design was one of my first attempts to design the infrastructure for the L plate app and plate. Ultimately, I've ended up changing it quite significantly but this was a very good starting point and through the design process and repeated evaluations. I was able to come to a better design.

This design uses 4G which does make it work off-line in isolation but it needs a mobile data connection and you also need an expensive pay per month monthly SIM card And it will mean the L plates heavier having a 4G data also consumes quite a lot of power so battery life is reduced. This design was not very good and was quite naive but it was a starting point for further design.

The design that I ended up going with was much more efficient as it does not require the extra component of the 4G SIM card but instead it uses Bluetooth and talk to the phone which can be easily charged and the GPS is no longer collected from the L plate but instead it is collected from the phone which means that the per unit cost is decreased and the manufacturing complexity is also decreased while having a more seamless user experience.

Because I had already done my first implementation. I could improve on the floors of my first ration with my second induration and remove the need for the GPS and 4G module which would reduce the cost and enable me to move more of the processing to the phone.

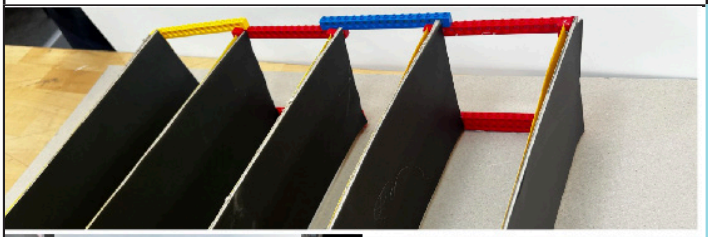
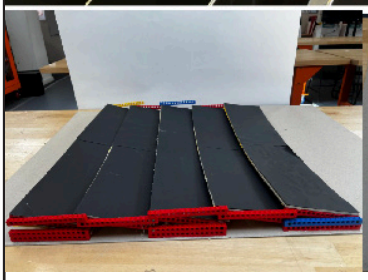
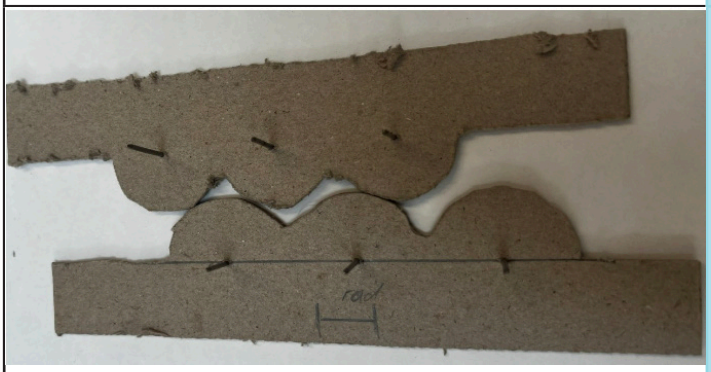
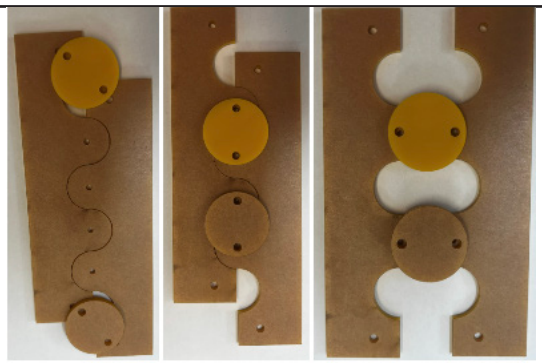
Project Development & Realisation

Factors Affecting Design

Factor	Evaluation

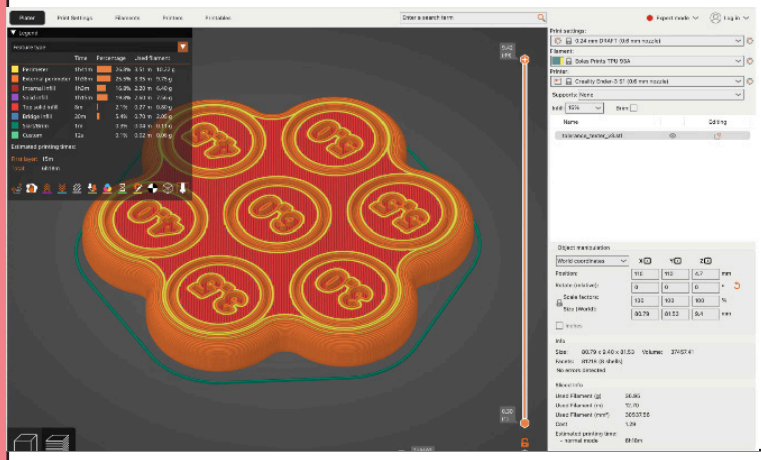
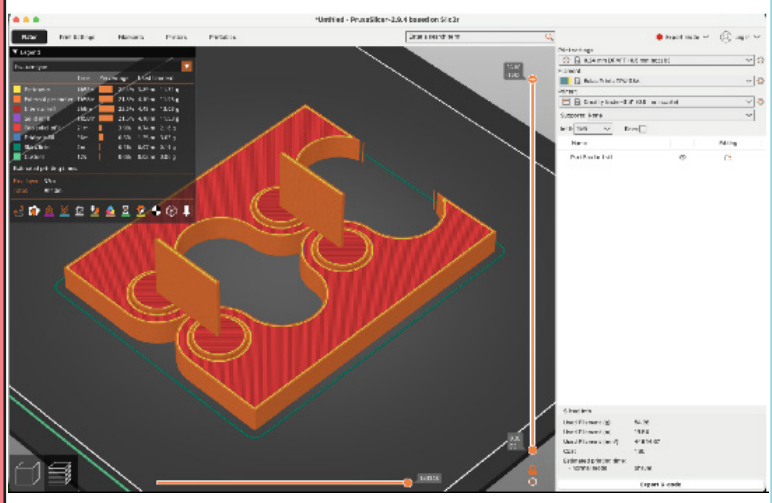
Project Development & Realisation

Prototypes

Prototype	Evaluation
  	Prototype 1: Template Steps 1. Used Lego blocks as a template to position the foundation blocks correctly 2. Hot glued one sheet of cardboard to the Lego bricks 3. Assembled the pins 4. Glued yellow and black paper to highlight how the mechanism moves and that it will work as an L plate This was the first test of the mechanism with the four bar linkage. This was an essential step to establish that this mechanism was a valid solution to the problem.
	Prototype 2: 1. I had a discussion with my mentor and we brainstormed ideas together until we came up with this idea of a mechanism 2. Next, I laid out cardboard and then cut it out 3. To simulate pins and joints I put nails through the centres of the holes 4. I then attached a plate to the nails to check if the mechanism could actuate fully
	Prototype 3:
	Prototype 4: I made the laser cut acrylic on Adobe illustrator and I adjusted the hole placement as per the previous one. I could reuse my same file but only making some small tweaks. I also changed from large circles to smaller ovals because this would allow me to see what was happening in the mechanism and better. This prototype taught me that my mechanism was completely valid although this particular mechanism was not optimal or optimised but it did work which was all it needed to do. There was lots of room for improvement but it was a foundation that would work.

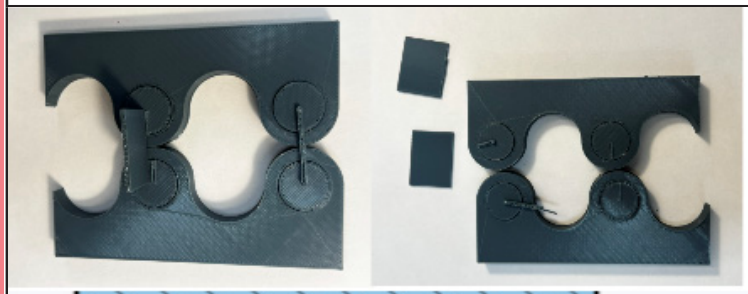
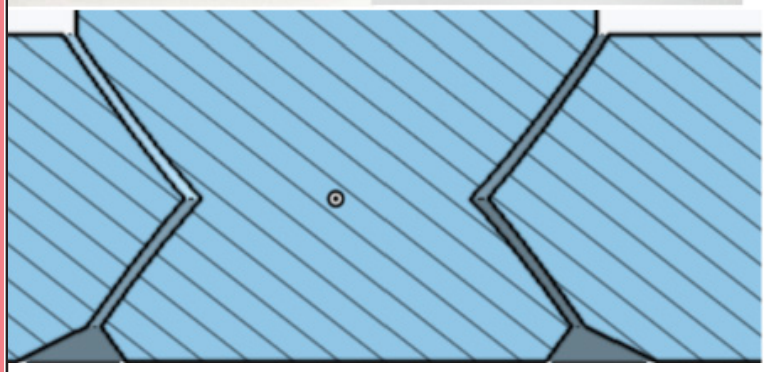
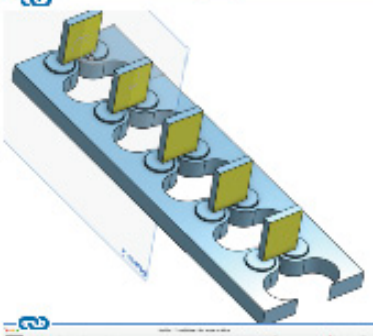
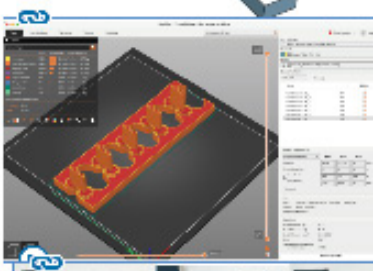
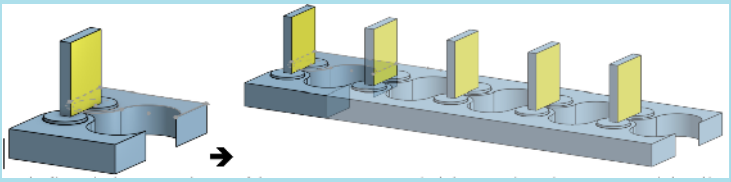
Project Development & Realisation

Prototypes

Prototype	Evaluation
 	Prototype 5: 1. Find the STL https://www.thingiverse.com/thing:3591280 it was on a site that is a Repository the 3D printed 2. Next I then imported into Prusa Slicer 3. I printed it on the school's 3D printers 4. The General process of putting on a 3D is to: 5. Load the filament 6. Preheat the bed and nozzle 7. Make sure the bed is clear 8. If necessary applying glue stick to the printer bed 9. Slice it with the default settings except a brim 10. Upload the sliced G code file onto a USB stick and begin the print 11. Click start 12. Monitor the first layer does not have any failures 13. Come back and take it off the bed 14. Then I took this off the printing bed and with my hands saw which ones of these I could move if I could move it it was a good enough tolerance if not it wasn't Ultimately the purpose of this was to be able to find out what is the tolerance of the 3D printer because I was doing print-in-place mechanisms and I need to know the gap that I need to leave so for my next prototype I can work out in CAD how thick it should be reducing the amount of iterations I have to do
	Prototype 6: I designed this particular part in Onshape I used a revolve to generate the dovetail pin I made sure I assigned the particular tolerance to a variable in the sketch I used so that I could easily modify it without needing to edit my sketch manually this is called parametric design and I used it throughout the entire design cycle exported the 3D model and then I printed it using the standard process

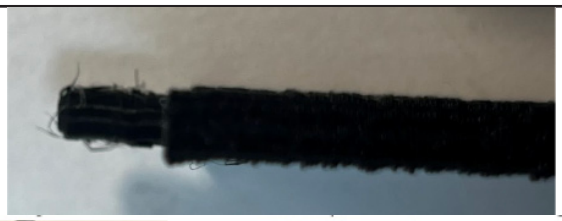
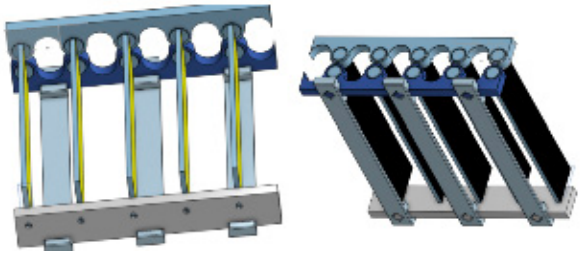
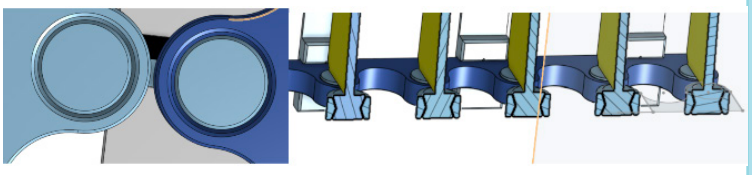
Project Development & Realisation

Prototypes

Prototype	Evaluation
 	Prototype 6 continued.. this allowed me to verify that the dovetail mechanism that I'm designing works as well as confirming that the design and shape work that was also additionally confirmed in the green acrylic 1. This is very useful because it allows me to as a small scale prototype refine the mechanism further knowing I have a working mechanism that only needs to be optimised from this point forward.
   	1. Given that you can quite clearly see a repeating pattern in the design that I wanted to achieve I was able to leverage the ability of CAD to do patterns so I started by making a single copy of what I wanted and then I duplicated it the number of times I wanted. 2. I defined the number of louvres as a variable so that later I could adjust the number of louvres I needed. This was very useful when testing 3, 4 and 5 different numbers of louvres.  3. I then printed it using the standard procedure Once I had printed this and verified that it worked, I could also test that a single one of these louvres could drive the entire mechanism. This allowed me to confirm that a single driving motor could work, which is the most efficient way of driving it. This is very beneficial because it confirms not only the mechanism but the way of activating it as well in future. I can start designing and optimising this mechanism for mass and area coverage as well as how to mount and drive it with a motor.

Project Development & Realisation

Prototypes

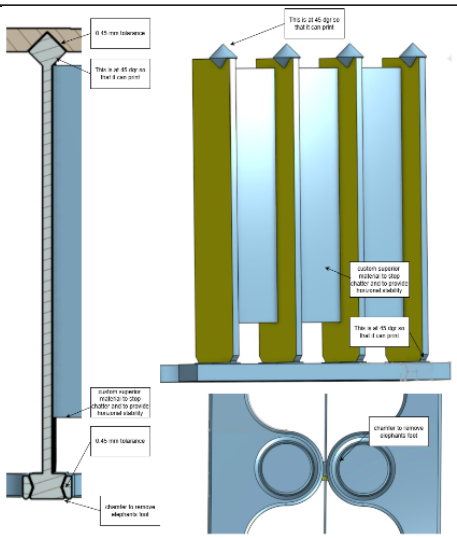
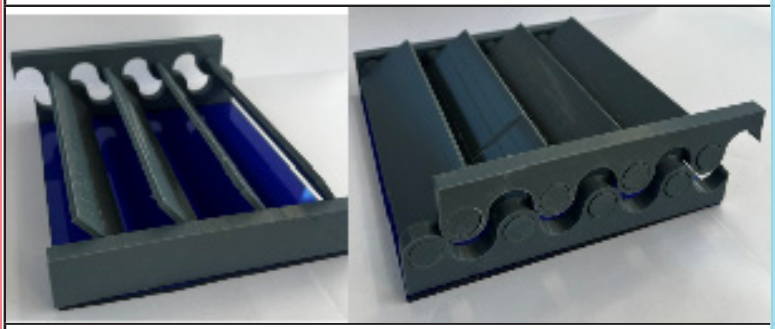
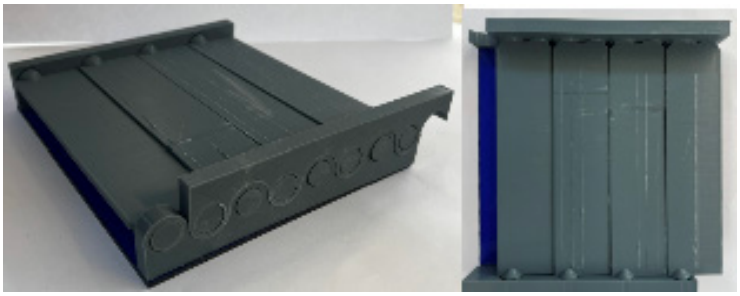
Prototype	Evaluation
	
	This image highlights the limitations due to chatter of the top pin
	These 3 images show the chatter this is why I added the custom support materials
	An overview of the mechanism
	
	Two images of the tolerances and the print in place mechanism

1. I was able to rapidly design to test whether this would work by not following parametric modelling procedures but instead just making it
2. The benefits of this is that it was faster but at the cost of having a reusable CAD model but I was able to revert at the end of this iteration back to the original model
3. One thing I did particularly was I made teardrop shaped holes. This allowed me to print at a better orientation as a teardrop hole will maintain tolerance as it won't sag
4. I also printed this using the standard procedure

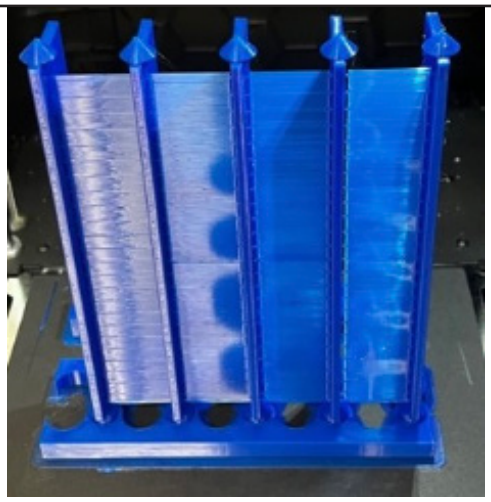
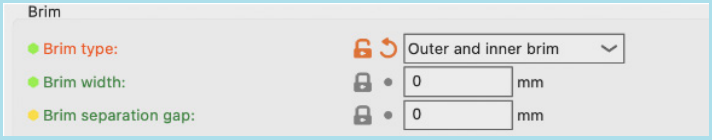
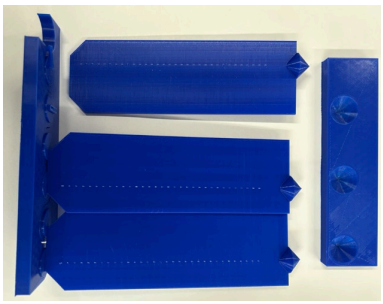
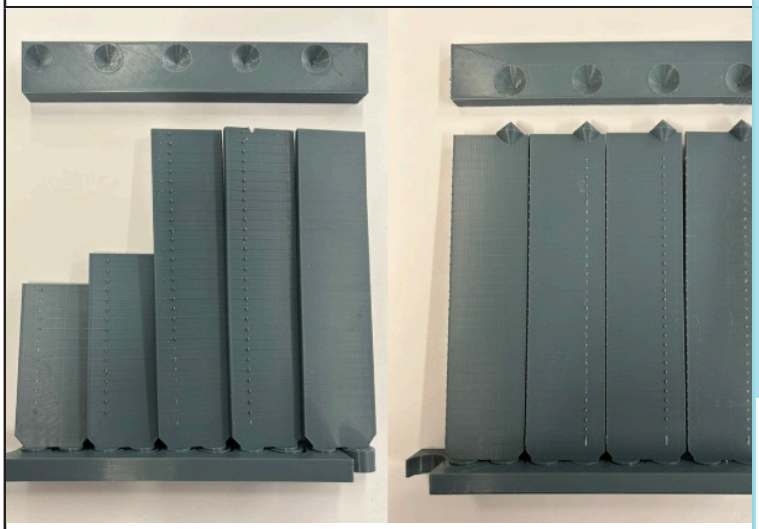
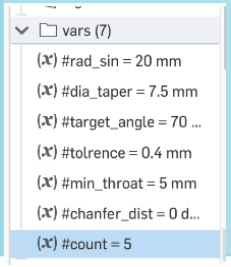
Ultimately this iteration of the prototype was somewhat unsuccessful but it did prove that the mechanism can work with an endpoint and full length levers but the downside was that the pins are really small and the bottom mounting method was not very reliable or strong as well as blocking vision so in future I need to work out some solution like a clear backing to join them as then it would not block vision

Project Development & Realisation

Prototypes

Prototype	Evaluation
	1. After researching about 3D printing optimisation I applied lots of the things I learnt as you can see in this annotation. 2. For example I added chamfers to remove elephant's foot. This is where the print squishes out and would cause the joint to adhere. I also added this large support material to stop them swaying. This is called design support. 3. The benefit of design support is if eventually I was to get this manufactured commercially. I wouldn't have to tune slicer settings on their end but I could simply get them to slice my model and it would work. 4. The reason I added 45° chamfers at the end is as for locating as well as maximising surface finish
	
	
	This print was incredibly instructive as it allowed me to determine that a lot of my optimisations were better as the mechanism that was less binding than the previous iteration as well as printing much more cleanly although the support material was very strongly adhered to so in the next iteration I should add preparations to allow it to easily detach. Despite me adding a chamfer for elephant's foot it was not large enough to properly mitigate it as I still had some evidence of elephant's foot so in the next prototype I will increase the amount of elephant's foot prevention I do

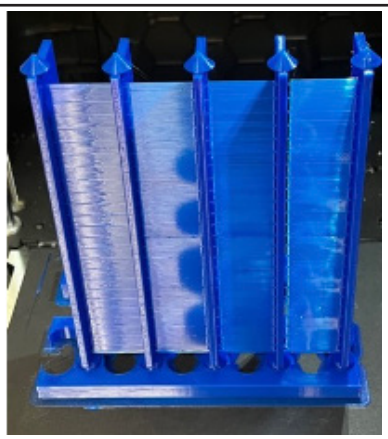
Project Development & Realisation

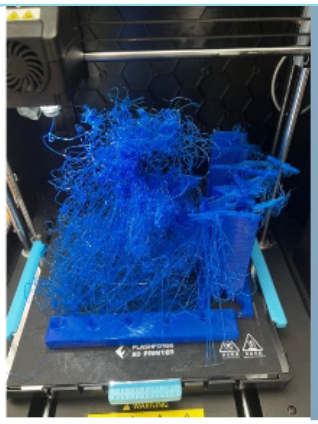
Prototype	Evaluation
	In this iteration I decided to add a brim. This would enable me to print it with better support and it did work, but it was almost impossible to remove and it damaged the mechanism. 1. I in CAD added perforations in the support material 2. In the slicer, I toggled the option for brim  3. As a side-effect of this I had the mechanism be bound up so to fix this I used a disc sander to grind away at the mechanism where it would not come away
	This wasn't good because it left damage to the mechanism, but the mechanism did function Going forward I will know that if I use a brim I should select an outer-only brim so it is easy to remove, otherwise I will struggle removing the brim.
	
	1. By adjusting the parametric variable for the count, I was able to generate three different louvres with different numbers of blades 2. 3, 4 and 5 successively this let me test what was the optimal number of blades for my prototype 3. I printed each of them using the standard procedure. I printed them all at the same time, though with different filaments. 

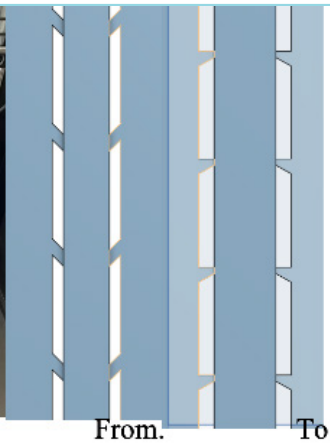
This let me determine the optimal number of blades to have on my mechanism as it was quite difficult deciding on the optimal number. As it is a compromise between the smaller I make it the smaller the pins have to be that are printed so I would like them to be as large as possible but then I've a side-effect of having to have thicker circles and larger blades which take up more space so to satisfy my design criteria I had to choose an optimum of these two that is why I decided at five as it seemed to satisfy all my requirements

Project Development & Realisation

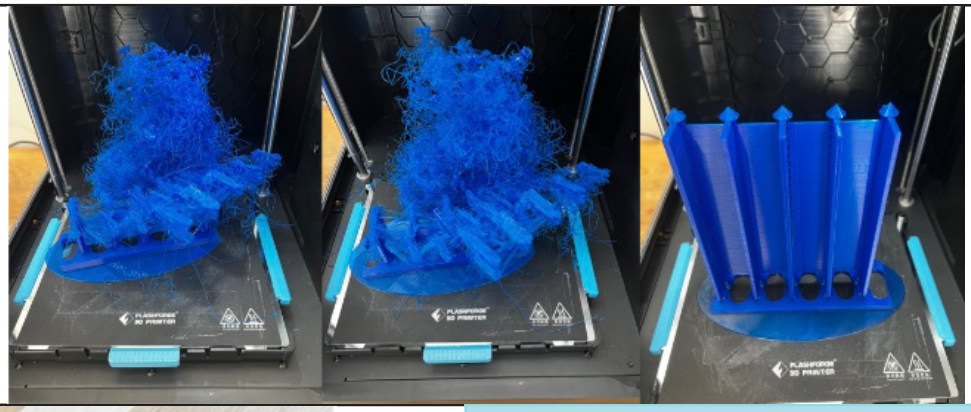
Prototype	Evaluation

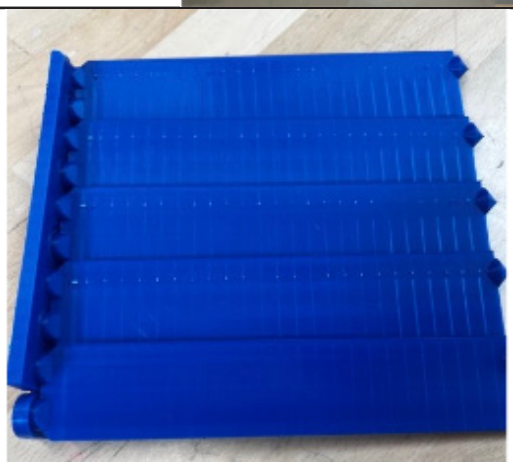






From. To





1. Onshape I changed the thickness the designed support from 1.414 mm to 0.400 mm so that it would be easier to remove I did this by editing a sketch

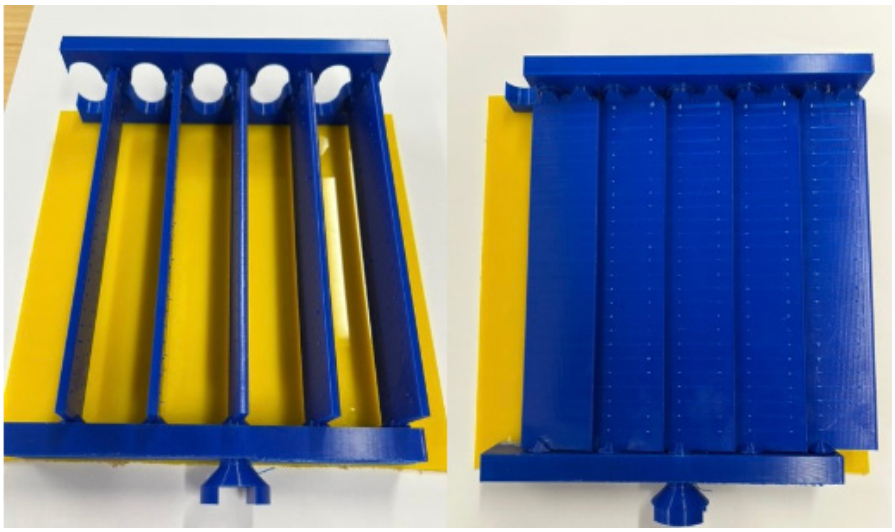
2. Then I printed it with a brim as seen in figure 1 this made it difficult for me to work out what my changes were as I forgot that I had made the small change in the thickness of the support material

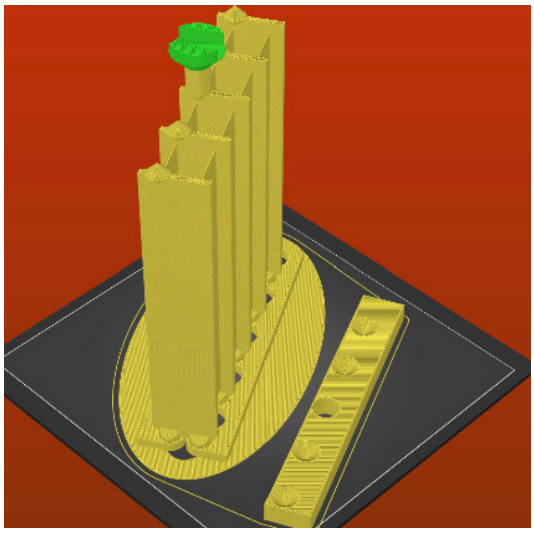
Basically what I did is I first printed one of them and then I printed it again with a better layer height and I did lots of very good quality bed prep with like glue stick and that also failed so then I printed it with a designed brim and that also didn't work. The issue was I kept on getting layer shifts and I didn't know why and it was actually because the louvres were existing as I can't believe her and they were snapping their limited area of the bed the printer so this fixed by me adding support material so that they couldn't leave it and it was actually one homogeneous piece so it would have enough surface area to stick to the bed and I did this across multiple iterations and I also added a brim in this step. This is a printing place support material which increases adhesion. I don't think it's necessary anymore like for the actual final one but I've kept it in there because it looks good, like I've designed things and it's complex.

This taught me that despite having thin support material which makes them very easy to remove You have to make a compromise about too strong or not strong enough and I made the wrong compromise by making it so thin that it came off and it didn't actually act as support material. That's why these prints kept on failing across three versions.

Project Development & Realisation

Prototype	Evaluation





1. To make this I didn't have to modify much. I modelled the actuator mount itself in CAD, and the only physical change I made was drilling a 13 mm hole for it to fit through. I used a drill rather than remodelling and reprinting the whole part because I already had one printed, so this was much faster

2. Then what I did was I pushed the print midway through. You can do this in the configuration in the slicer settings so that the print will automatically pause at the correct layer height. You can see this in the image with the green and yellow a slicer screen shot. The green is where it all pauses so you can change colours.

3. This allowed me to place the bar on top of the louvres in place and then I could resume it and print the drive mechanism

4. Once I finished the print, I took it off and glued it to a base and I also lubricated the hinges

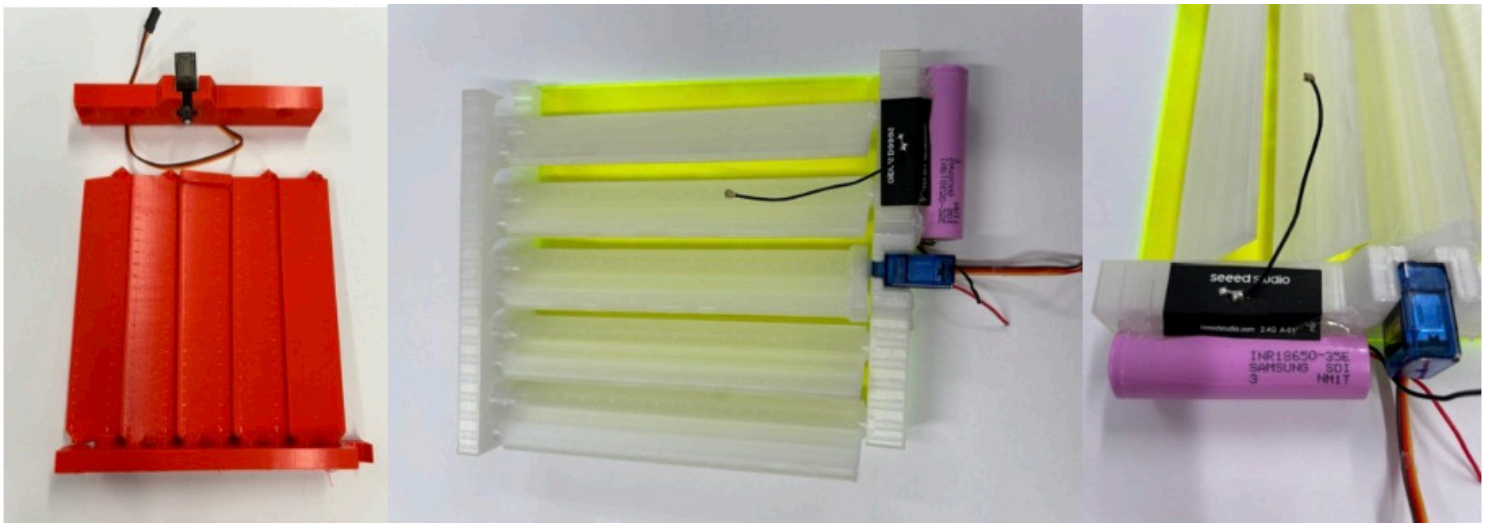
This let me establish whether it was possible to drive this mechanism from one single point as this is how I intended to design and make a servo mechanism for it to go into. I could also do load testing with the servo to see if the servo was powerful enough to turn it and it was, which means I can use a small SG90 (this is a typical small servo driven by a PWM [pulse width modulation which means a series of pulses which encode digitally the position that the closed loop controller must try and drive to] signal from a microcontroller). Having a servo means that I can be more reliable and precise as a servo is a closed loop controller which means it knows position so if the gears slip or a kid holds it down it will still try and move once pressure is released instead of losing position as the case with 3D printing stepper motors.

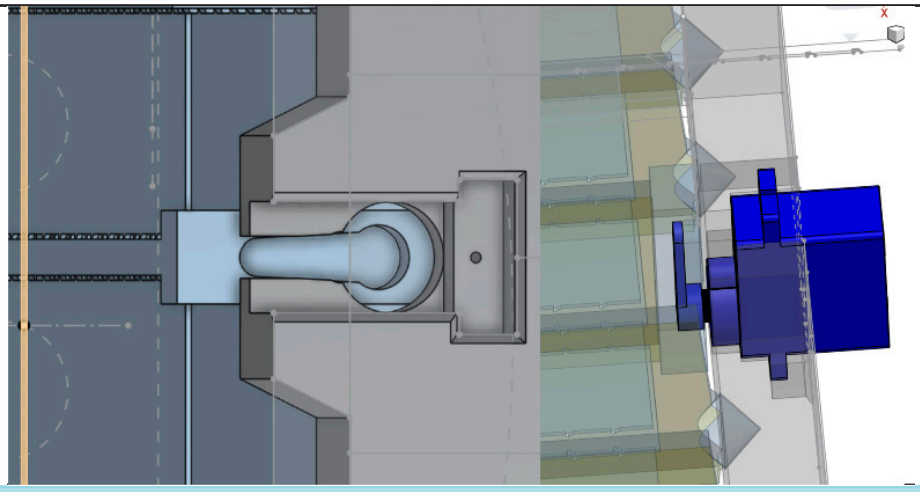
In addition this pausing the print and resuming is how I'm going to have the multicoloured L plates and it was a good verification of this manufacturing technique

At this point, I'm quite happy with a louvre mechanism so far and I would like to move onto now prototyping the electronic platform

Project Development & Realisation

Prototype	Evaluation





1. The first step whenever you're modelling something in CAD to work around an existing part is to always import or make a model of the object you're trying to make a mount for or at least a rough approximation of the parts that you need to interface with

2. This meant that I imported a 3D model of the SG90 servo with a horn that I was planning on using as you can see in the image.

3. Then using a series of extrudes and sketches I cut away the material. In these sketches. I included an offset with my tolerance dimension. This is a globally defined dimension which is set to 0.25 mm. This is the tolerance of the 3D printer and it means that when I insert things it'll have enough clearance to physically fit.

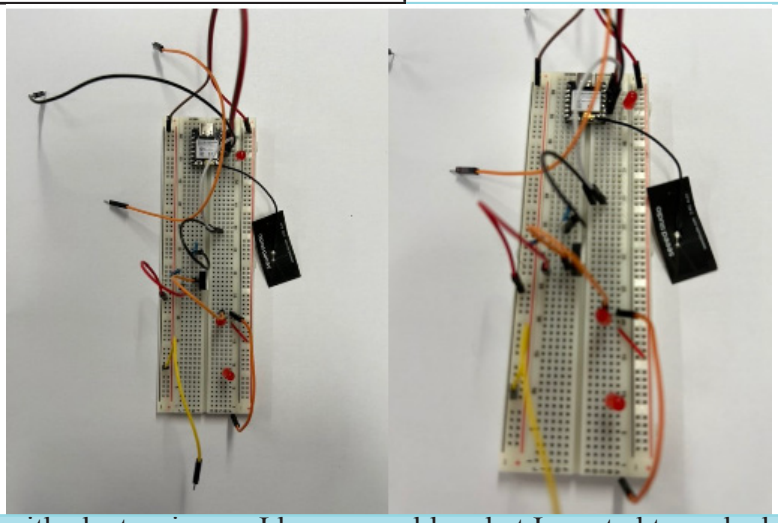
4. I decided the second servo arm mount is not necessary so I will be cutting that off so it can fit and reduce space which is one of my success criteria

The red one Is printed in PLA whereas the white one is printed in PETG. PETG is known for better UV resistance and also being tougher. While PLA is much faster to print and good for rapid prototyping it does not handle UV well so that is why I printed in PETG so that I could get a feel for what that material is like. Going forward I will continue to use PETG when necessary but if there is no PETG in the DT building I will use PLA as it's fine for a mechanical prototype but does not necessarily show the final product or the same printing tolerances.

I've also learnt that my servo is strong enough to drive the mechanism. This means going forward that I can use a servo and continue designing and pursuing this design route. Next. I need to work out some way of mounting the battery which is not just with hot glue as currently it's crude.

Project Development & Realisation

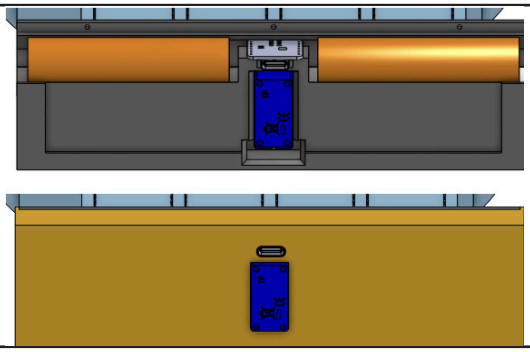
Prototype	Evaluation



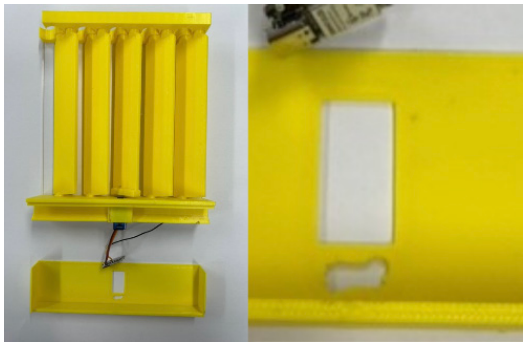
I have had some experience with electronics, so I knew roughly what I wanted to make but I was in contact with my mentor and ended up having a two hour conversation where we planned out the electronics of the board. This was our first version and prototype. It worked, but I decided not to keep the mosfet (it is essentially an electronically controlled switch, this particular mosfet defaulted to off and only turned on when a voltage was applied to it). The idea was that it would let me completely disconnect the servo so it could not draw any power at all, and I could also put the ESP32 into deep sleep, which would have given me much better battery life. In the end I decided it was not worth it, as the mosfet added cost and complexity to the board for a fairly small gain.

But instead I have decided to go down a route where I will just send a PWM signal of nothing to the servo which means it will not try and actively drive and finish the control loop which will have 95% of the battery savings while reducing weight and cost the end user.

Prototype	Evaluation

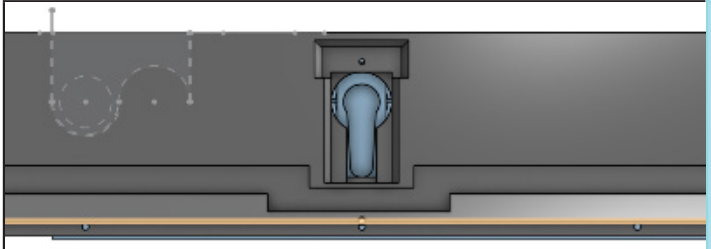
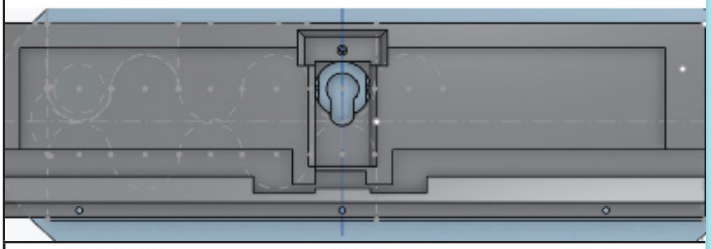
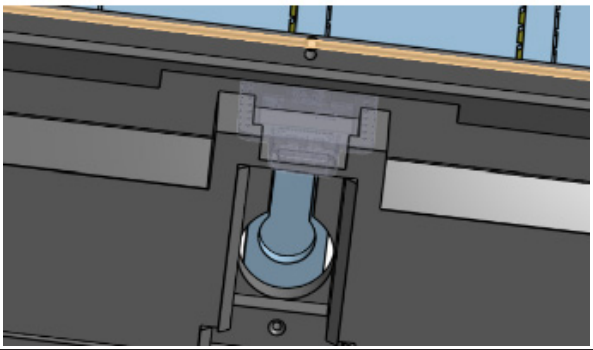
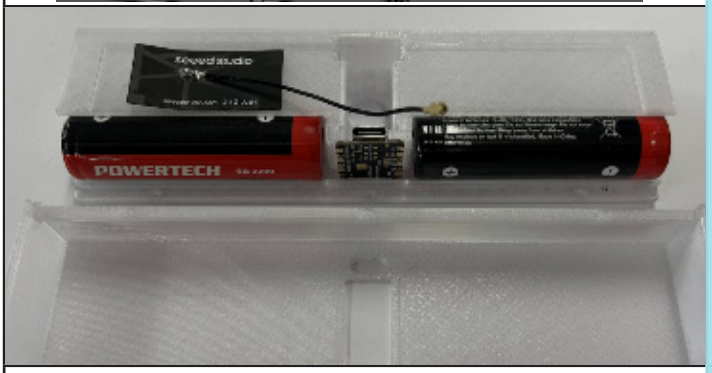
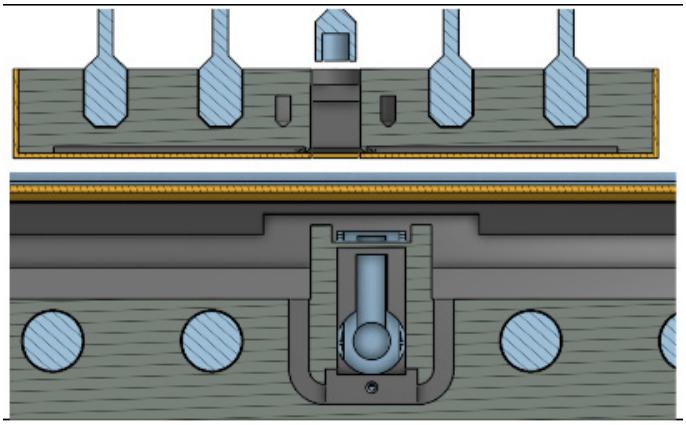


1. Following the established design pattern of placing the components first and then modelling around them I placed the batteries in a spot that I liked
2. To mount the batteries I made a V groove. This was a easy model and there was room to route the wire underneath and see it was also had a snap fit for the batteries would stay in place.
3. For mounting the ESP32 I had made a decision to mount it with the PCB labelled down. This allows the antenna exposed and the USB cable to be accessible but because the mounting was not repeatable and because of the heat generated this led to the ESP32C3 (this is a low power micro controller with Bluetooth and Wi-Fi functionality and an integrated battery charging module)

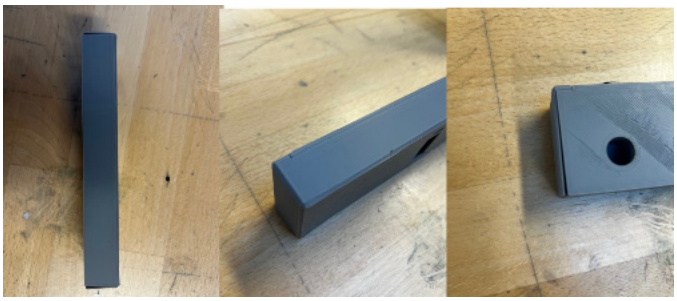


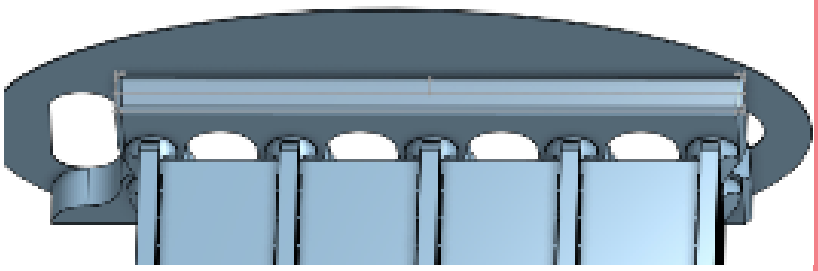
Things I didn't consider and should have : I didn't consider the wire routing as wires have a physical size and I didn't consider how they should be routed and this meant that the case would not fit on. I also did not make the hole correct spot or the spot was correct for the CAD model but since the ESP32 could move it didn't end up being in the correct spot so I had to use a drill to try and widen it which did not work out very well. I also didn't consider that there was no way to access the charging module pins underneath the ESP32 as they were hidden by the plastic. These various mistakes are what happens when you're on the first or second iteration of your electronics module and there's a significant amount of moving parts and complexities that you have to keep in your head and you make mistakes. That's okay because this is why we prototype. I needed to make sure that I include proper wire routing (this is foreshadowing) and also mount the servo correctly

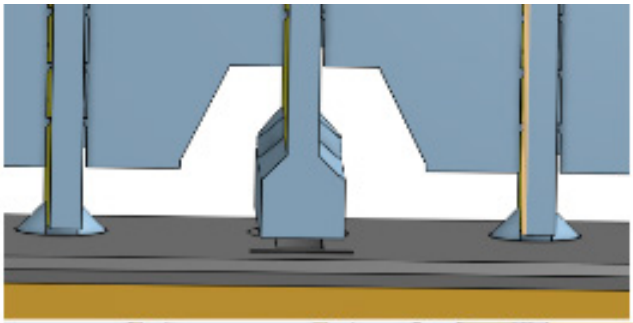
Project Development & Realisation

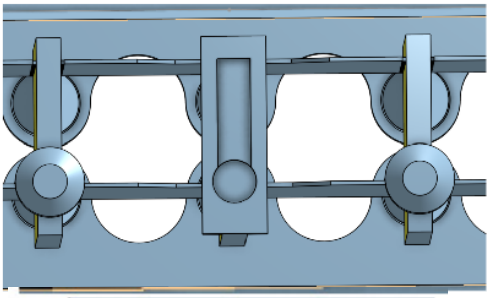
Prototype	Evaluation
   	1. As wires are very difficult to model I also made an error in this one. I accounted for the wires on the battery holder but I didn't account for the thickness of the wires going from the servo motor to the ESP32. I also made fix mounting on the top I did this by turning the ESP upside down and then doing a series of sketches and extrudes to make this boss that the ESP could sit on the back of the ESP was constrained by the back plate and the sides around the CPU and the ESP and the USB port were constrained by the two walls meaning that it would be in the same spot every time. I also adjusted the location on the outside case accordingly. 2. This let me make the ESP be in a repeatable spot every time which allowed me to design a correct size hole Going forward, I need to make a solution for routing the wires correctly This new layout worked really well although there was an issue with there being no way to access the boot button on the ESP32 the boot button is needed to reset the ESP32 when programming it with code. But other than that this setup was very effective despite that being no way of routing wires from the servo to the ESP32 that did not the case to bulge. You can see in the photo above that I have used a drill to test a different way of routing the wires. My first approach did not work very well but my second approach routing them through the body worked very well. That is what I decided on going with for the final wire routing solution from the servo to the ESP32
	The reason that the channel on the servo wire routing Is curved is the orientation that print out and it's best to have a 45° angle. This is what I learnt in my research and a 45° angle or a horizontal 90° angle that is the optimal because if it is a horizontal, the printer can't stretch the filament from one pole to the other like a bridge and if it is 45° it is a general rule of thumb number about how steep of an angle it can have before it starts sagging so I used a combination of 45° angles and a bridge, which is the name for the 90° horizontal. This let me effectively route wires through a channel without the risk of having filament in the way as it would be difficult to clear out. I chose to make the channel on both sides instead of one as there was no reason not to if I make it on one side and it turns out that for some reason the batteries sit better if the wire goes the other way then I'm in a bad issue, but if I have it in both ways, not only makes it stronger because there's more material there because remember infill is not as strong as but that it will sit better depending on the particular case

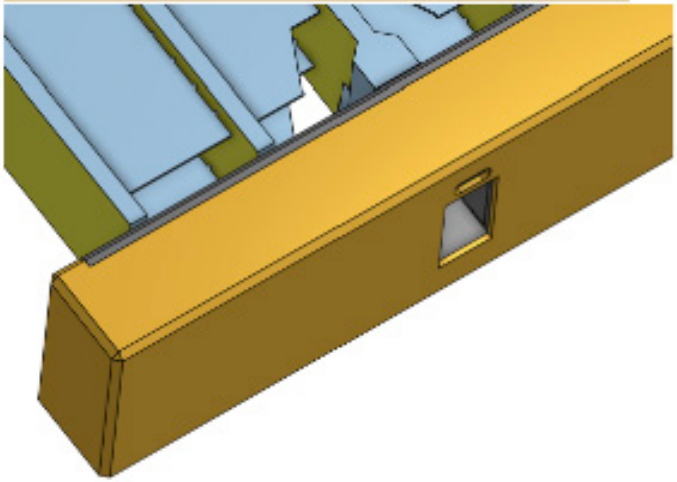
Project Development & Realisation

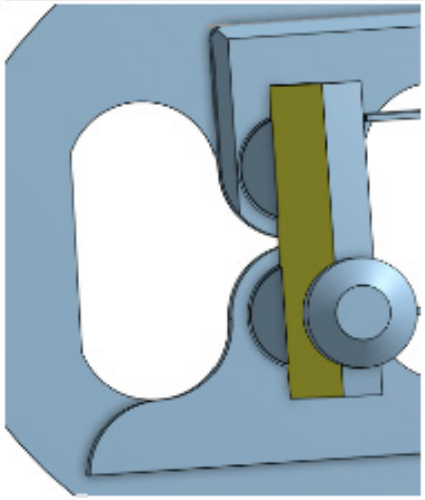


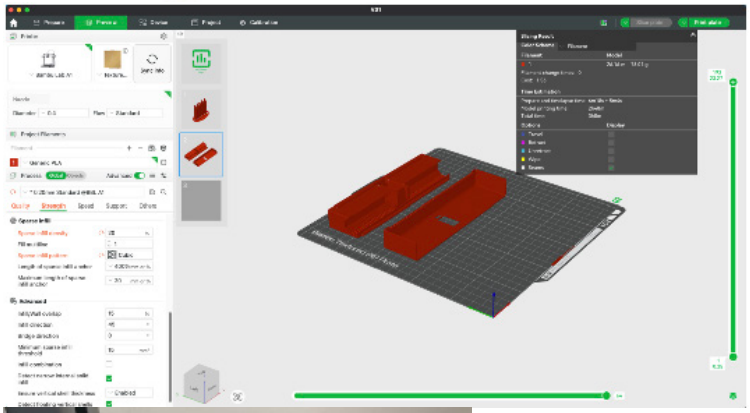


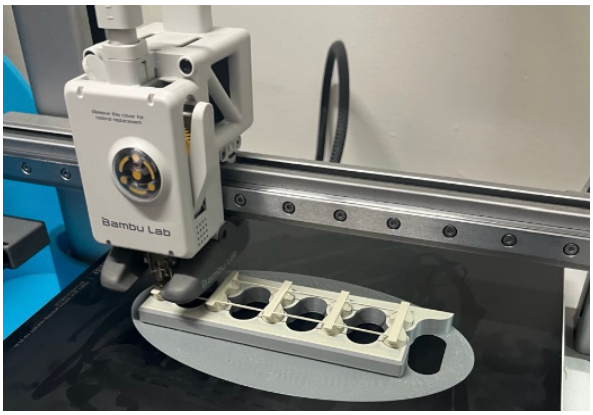


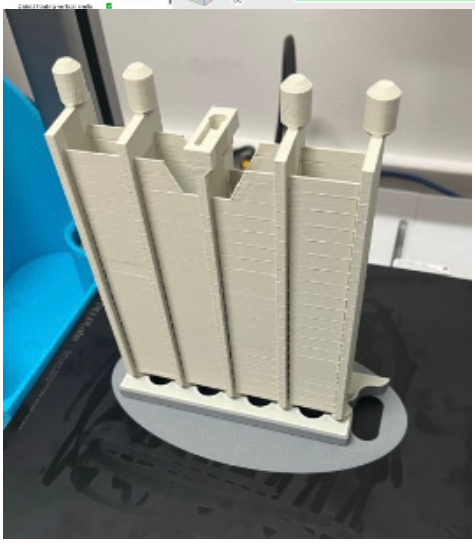


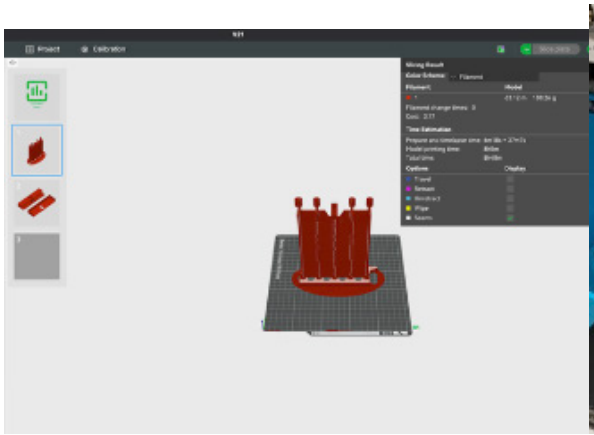


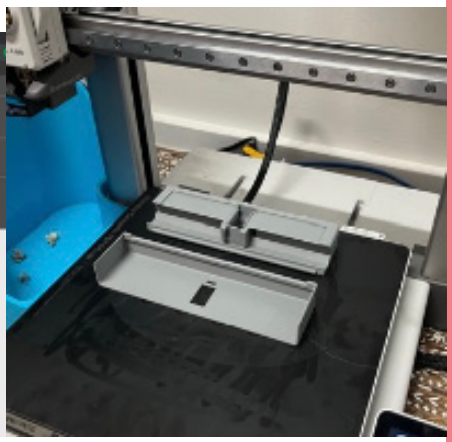






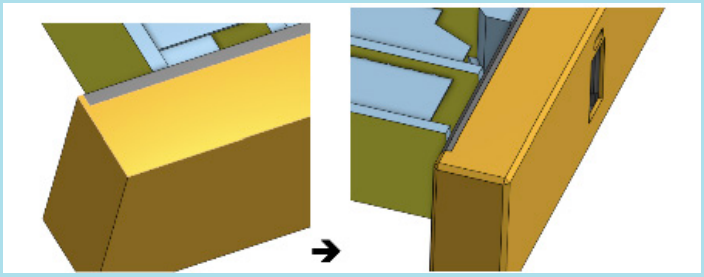






Project Development & Realisation

1. I wanted To increase the hole where the servo horn mounts as in previous iteration I had to use a drill and a heat gun to mount the servo horn into the hole so to do this I used an extrude but I also needed to chamfer for the sides of the Horn mount as it would mean that it would be stronger and not have a chance to fail as it's printing in thin air as a previous iterations did
2. I needed to remove material from my support material so that it could accommodate the larger horn. I did this with a simple sketch and an extrude the whole way through.
3. I wanted to make the edge of chamfered to improve the safety of my product to do this I needed to often chamfer manually as I wanted to keep the brim place.
4. You can see highlighted and figure 2 that the brim looks like it's in place and weirdly fitting this is because instead of properly modifying the parametric design I simply used an extrude to cut away and adjust the shape. This is risky but it is not worth modifying the original brim as there is a small cost in filament but a large cost wasted time fixing the parametric document so it would be easier just to cut away.
5. Because the corners were originally mounting in the edge



7. You can see how I move the location of the corner to be inwards so that there would be a solid material to make the chamfer on otherwise you would end up with a gap as a tolerances on chamfers are slightly worse than normal and this would look ugly and could cut someone but because I moved it inwards now you have a solid piece of material you're chamfering which both increases its strength and also improves the aesthetics

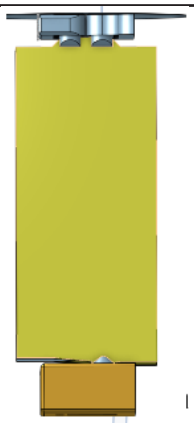
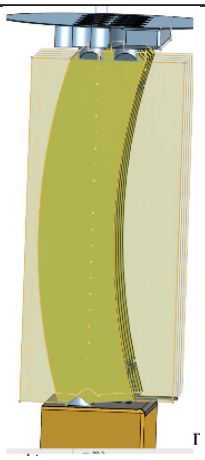
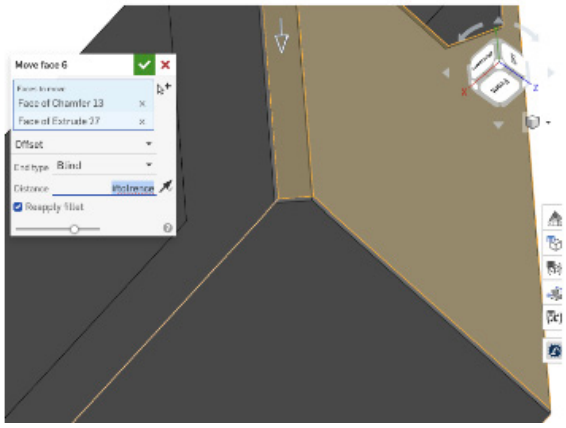




8. You can see this top corner. I did not include any tolerance. This means that the top corner acts as a focal point and it will pry open the bottom of this bar. Next time or for the next iteration. I need to increase the chamfer amount on the inside as I do on the outside so that it is a relief so that it doesn't pry open the lip

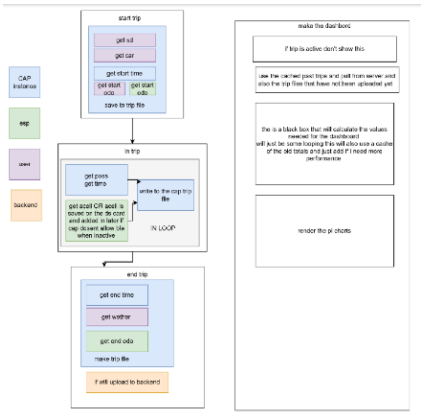
Overall the chamfers look really nice and clean but there needs to be improvement with refining tolerance in the case but ultimately these are small refinements that make a product high-quality and good design and it's important to get this right.

Project Development & Realisation

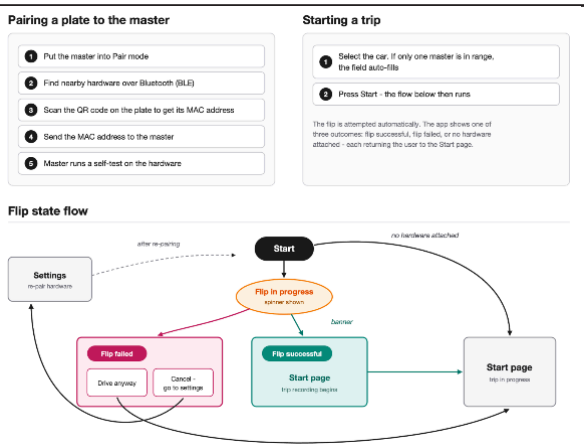
Prototype	Evaluation
	I extruded this to be like this then I can cut them to make it curved
 	Then I used a sketch and a extrude to cut away the luvers in the right way Then I used a file to remove the gap I had to remove material that was both in the shampoo and on the side face so once I've done that I removed that material in CAD for the next iteration

Project Development & Realisation

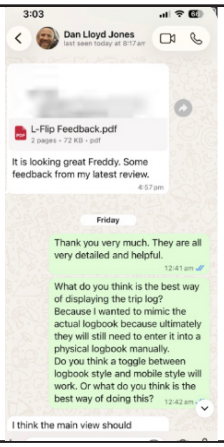
App Development



This is my initial design for the UI but ultimately I decided not to go with it as I spent quite a considerable amount of time user testing to try and optimise the UI for efficiency and intuitiveness.
Due to my testing, I was able to provide improvement to this design because the new designs were more intuitve to use and more aesthetically pleasing.



This float chart is quite efficiently conveys how the pairing process works in edge cases for example when the plate will not connect to the phone and how it handles this gracefully



Communicating with a Mentor he is a professional software developer of mobile apps

Project Development & Realisation

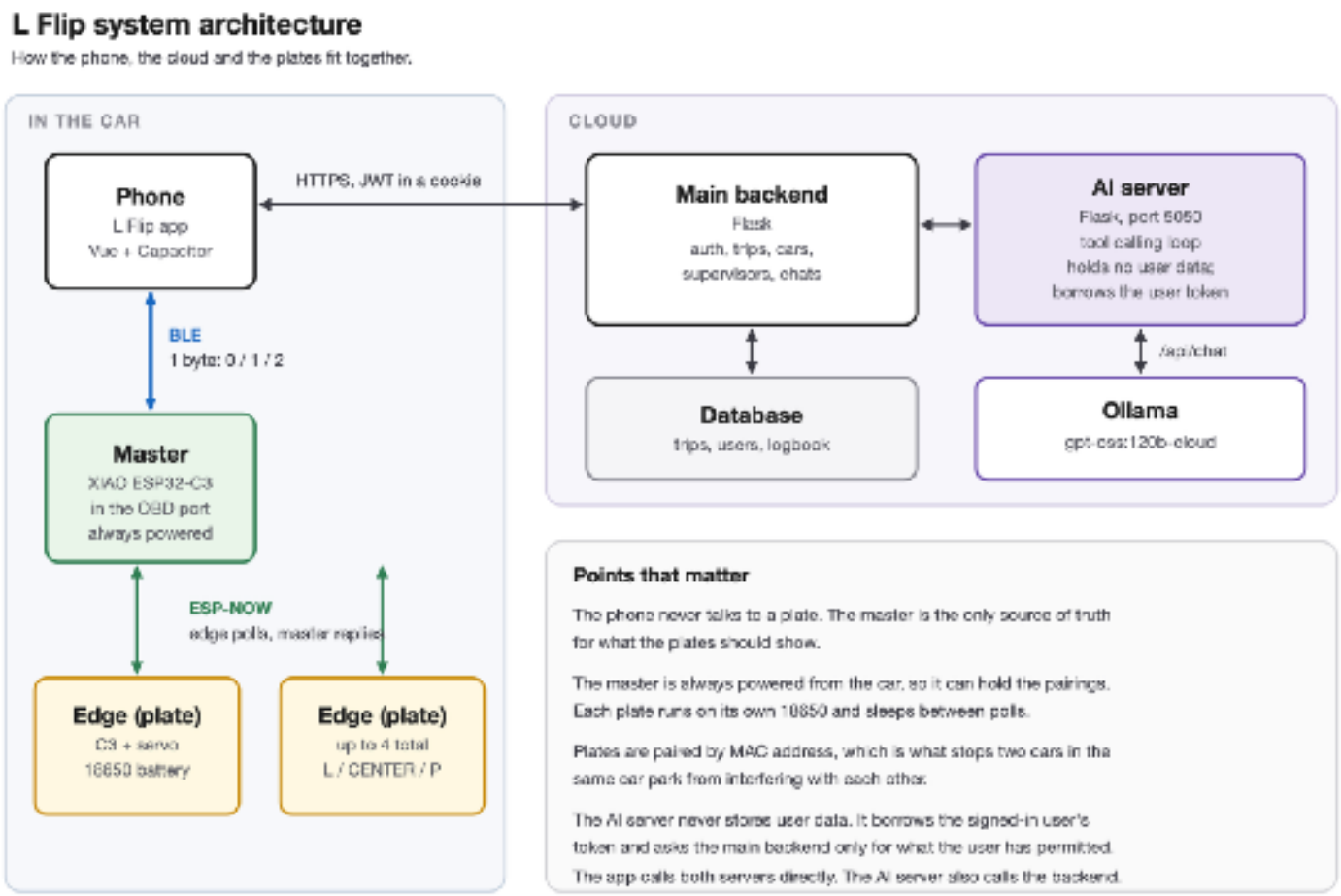
App Development

Intended result	Actual result	Result	Action taken
Taken to dashboard	Taken to dashboard, you were routed through the login screen but this is so that I don't have to duplicate logic	PASS	NA
Taken to login screen without a JWT or cookie	I got taken to the onboarding screen	FAIL	Remove the cookie
Rejected by a generic but helpful error message	I got good error messages that did not give away the back-end infrastructure but were clear enough to understand	PASS	NA
Returns personalised information about your dives	I got a "Sorry, I couldn't produce an answer for that. Please try again." I got this message	PASS	The reason this did not work the first time is because I made the chat and then the tool separately
Verify that the last AI chats are gone	They were not removed	FAIL	Execute a force reload; this is an easy fix. Open further investigation — there is another cause
Verify that there is some sort of error message that doesn't allow this	Got a helpful error message	PASS	NA
You log in successfully	It worked but I got the onboarding pages. This will be annoying	PASS	Make the onboarding only come up if flag exists, not if flag doesn't exist. Fix in progress
I would expect that all the text is readable and the themes match up with no excessive amounts of light	I found a few text boxes and themes that were not properly in dark mode	FAIL	Update my CSS and theming as well as my colour palette to work with dark mode
The expected result is in the first case to get trip information and other specific information about the user	My server hardening worked very well, but the AI tool fabricated data (see 1) for the output	PASS	The fix is to harden the system prompt
I'll be able to see it in the GPS map. GPS points will still be logged despite the screen being off		not run	
Get a pop-up or other notification from the LM that you're out of tokens	This was really bad design on my behalf. All you got was a pop-up saying please check your connection	PASS	I was HTML-escaping the content twice which meant that I was getting things like " in the output
The master and slave will connect, then they will seamlessly toggle with an intuitive UI	This works as expected	PASS	NA
The app icon should be a bright yellow with a black L	The app was an olive green-yellow. This looked bad	FAIL	So I changed it to a bright yellow

I conducted something called acceptance testing. This is where you will test the functionality of your app and after testing the functionality of your app you need to verify that it did what you expected to do. If it doesn't you have a problem and you need to fix it but if it does work it pass and you know you're verified the app works. This is different to something called unit testing which is where you test each piece of code individually.

Because I did this thorough acceptance testing, it allowed me to verify that my app worked and was able to be navigating without any bug bugs giving me great amounts of peace of mind that I could keep on pursuing the design of my app

Project Development & Realisation



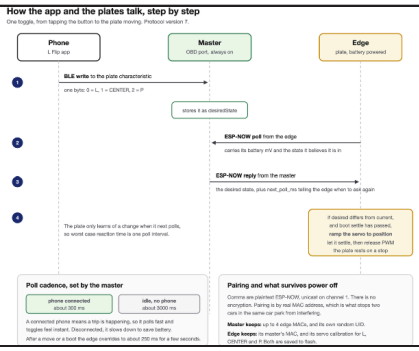
This is how my Software works

There are lots of components that need to interact with each other and this means that it is quite complicated so being able to make these visualisations was really helpful in me understanding how my own system works so that I could find bugs in my code.

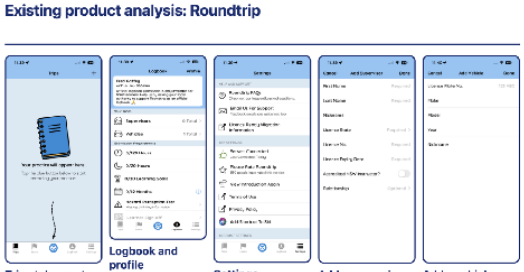
This architecture has been validated by testing and this gives me confidence to move further onwards. It follows industry best practices for security as well as having isolation between the AI server in the main back end as AI products can often deal or lead to security vulnerabilities in the app. That is why I have it isolated.

Project Development & Realisation

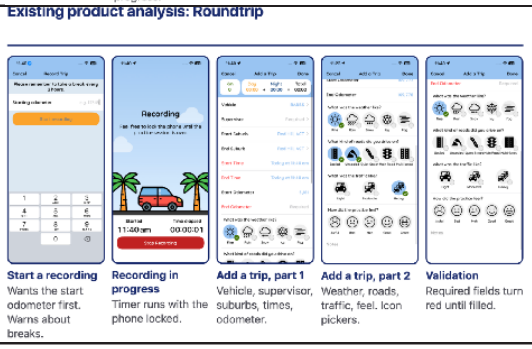
App Development



This is a diagram outlining how I am intending to have my network protocols to work. I design this so that the user would not have to think about how it works, but it was intuitive.
Also, so that if we reset with no maintenance.This proved to work quite well.



This is me analysing my competitors app and their app works quite well. Lacks a lot of the advanced features that As mine is designed to be integrated with hardware and log hours very fast with minimal friction whereas. But I was able to gain meaningful insight from what they have and it helped me work out how I was going to design mine.



The way that implemented weather selection UI helped me choose to use a drop-down instead of a text box. Because a text box means that it is hard to search in categorise whereas clicking a drop-down or button is much more efficient.

Project Development & Realisation

Identification and Justification of Ideas and Resources

Tools

Project Development & Realisation

Identification and Justification of Ideas and Resources

Materials

Project Development & Realisation

Identification and Justification of Ideas and Resources

Techniques

Project Development & Realisation

Use of Communication and Presentation Techniques

	On shape is a web-based 3D card platform and it allowed me to rapidly prototype my designs. Another feature of shape is that it has virgin histories so every edit I make has a saved checkpoint and you can revert back to previous versions. In software we have something called git this is a version control system that uses branching This makes it very familiar for me. In design also let me generate cross-sectional views of my 3D models which were very useful in conveying how the internal geometry worked
	InDesign helped me re-organise and format the folio so it is in line with the marking criteria for it to be succinct and well organised. I was able to colour code each section and make it more aesthetically pleasing
	Microsoft Word let me draft and spellcheck my words in writing before I submitted. It also gave me the ability to share with my teacher to gain feedback. This was useful for improving my portfolio and communicating articulately and succinctly.
	I used Excel to easily construct a well formatted time action plan, that I could print out and easily write down the steps and how long it took me to complete. I could also easily evaluate these steps.
	Adobe illustrator let me design and test my very first iteration on the laser cutter.
	Draw Io let me generate beautiful diagrams that could clearly communicate the flow of complex unintuitive concepts in programming. This let me communicate visually instead of through code examples or pseudocode which despite being easier to read them real code it is still difficult

These communication tools let me effectively convey my ideas without these. I would have to write dense paragraphs of wordy complicated technical text and this would struggle to communicate effectivelyWhat I wanted to say but instead I was able to use tools like Onshape especially to show things like internal geometry without having to physically cut open a model and take a photo of it

Project Evaluation

Evaluation of Functional and Aesthetic Aspects of Design

Project Evaluation

Evaluation of the Projects Impact on the Individual, Society and Enviornment

Project Evaluation

Relationship of the final product, system or enviornment to the project proposal